

Hedge Overlay Recommendation

model_v4 — "up-shock + vol-surge" 1-day semis hedge

Standalone overlay (src/hedge_overlay.py); the model is NOT modified.

THE RULE (decide at close T, hold T+1 only, exit next close)

FIRE if: QQQ 1-day return $\geq +2\%$ AND $z(\text{SPY 5d realized vol, 63d}) \geq 1.0$

THEN buy: SOXS (long -3x inverse-semis ETF -- no shorting), sized INVERSE-VOL

to ~14% of CURRENT position market value (mark-to-mkt, ex-cash);

hedge \$-vol = 50% of book \$-vol, scales with exposure & vol $\Rightarrow$ ~\$14k per \$100k exposure.

WHAT IT TARGETS

- Sharp up-days that occur while volatility is already elevated -- a combination that tends to mean-revert (pull back) the next session.
- No single factor works alone (a QQQ up-day alone BOUNCES); the edge is the INTERACTION of up-shock x elevated vol.
- Best supporting signal (sensitivity): 5-day realized-vol z-score. Relative vol beats absolute level / surge-ratio; the 5d window beats 1d and 20d.

QUANTIFIED VALUE (\$100k book, SOXS inverse-vol ~14% avg, 68 fires)

1-DAY hold (THE RULE): net \$+26,846 [wins \$+47,962 / losses \$-21,116, 57% win]

if held 5D: \$+13,032 if held 20D: \$-117,571 $\rightarrow$ the 1-day exit IS the edge

best saves: 2020-03-13 \$+7,369 (book -9.0%) - 2026-06-16 \$+2,034 (book -3.8%)

RISK-ADJUSTED (SOXS inverse-vol)

FULL: dSharpe +0.11 sleeve PnL +26.8% worst on-day -9.0% $\rightarrow$ -4.2%

OOS (≥ 2024): dSharpe +0.08 sleeve +5.2%

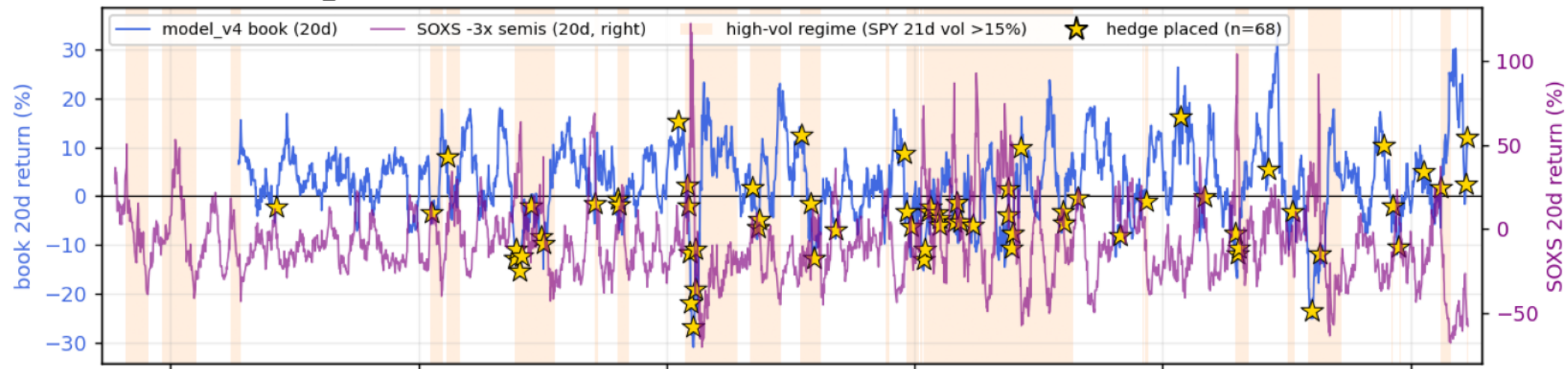
Walk-forward: 7/10 active years dSharpe >0 ; gains concentrated in vol-spike years

CAVEATS

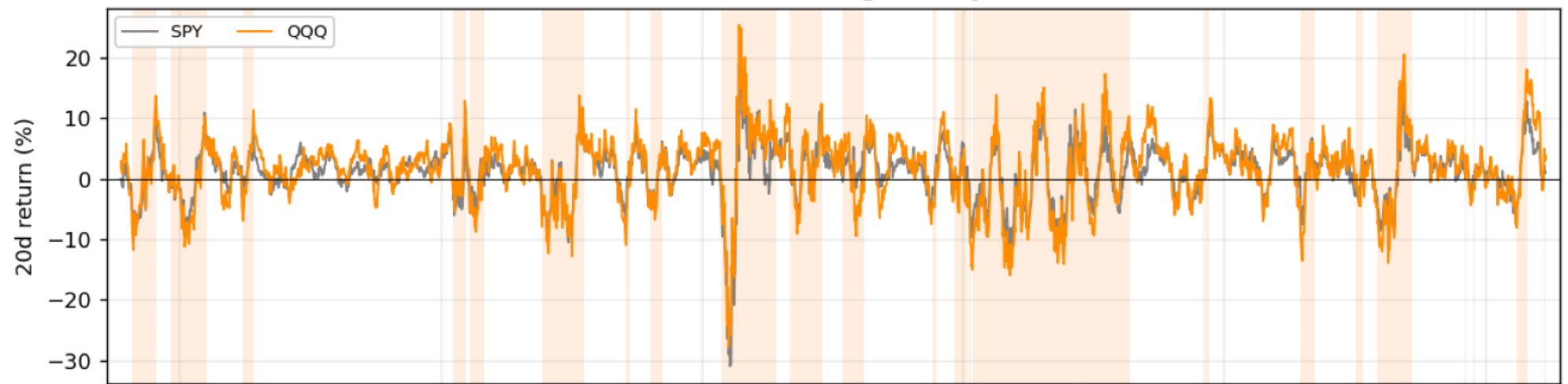
- A Sharpe-adder + single-day-blow softener, NOT an all-time-max-drawdown reducer.
- Edge concentrates in vol spikes; marginal in calm years; slightly -ve in 2022 bear.
- Costs use a flat 10bp; SOXS real spread / expense / 1-day -3x decay may be higher.
- Paper / recommendation only. Re-validate after any model or universe change.

SOXS hedge, inverse-vol sized (~14% avg notional) on \$100k book – PnL by hold horizon (n=68 fires)
1-DAY (the rule): net \$+26,846 [wins \$+47,962 / losses \$-21,116, 57% win]
5-DAY: \$+13,032 20-DAY: \$-117,571 -> strict 1-day exit is what saves \$
best 1D saves: 2020-03-13 \$+7,369 (book -9.0%) - 2026-06-16 \$+2,034 (book -3.8%)

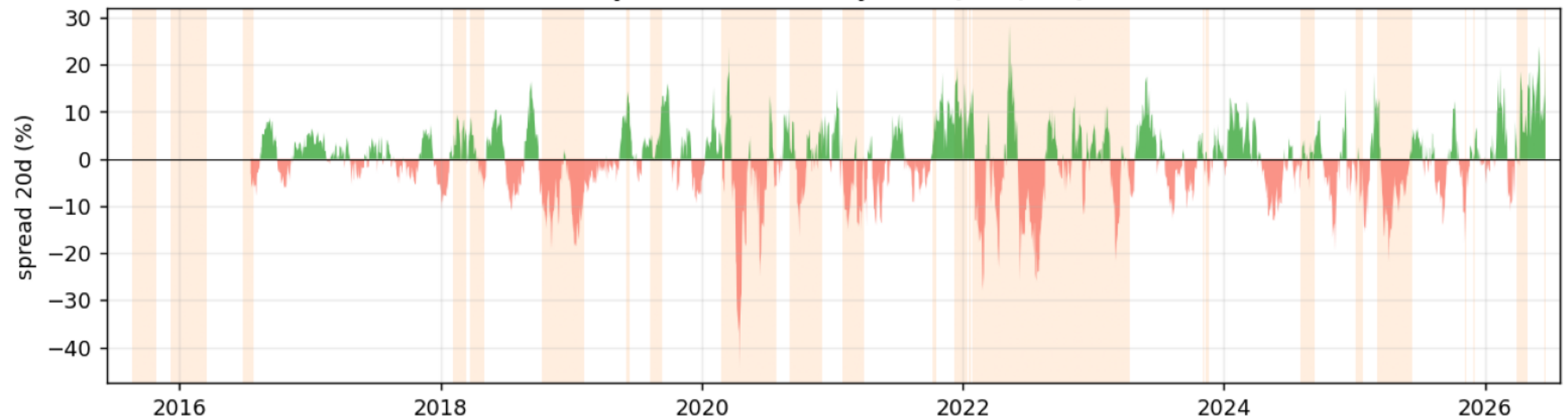
model_v4 portfolio vs SOXS hedge — ★ = hedge placed (up-shock + vol-surge)



SPY / QQQ — trailing 20-day return



Top10–Bottom10 spread (20d, fill)



Backtest — windows & yearly walk-forward

Default: SOXS (long -3x) sized inverse-vol (hedge=50%% of book risk), trigger $qqq_1d \geq 2\%$ & $z(spy_vol_5d) \geq 1$. dSh = hedged-minus-book Sharpe; dDD = max-DD change (+ = better).

WINDOW SUMMARY

window	days	on		bookSh	hedSh	dSh		bookDD	dDD		onBook%	paid%	sleeve%
FULL	2513	68		+1.24	+1.36	+0.11		-39.5%	-0.6%		-0.44	57	+26.85%
COVID	1561	47		+1.22	+1.26	+0.04		-39.5%	-0.6%		0.07	51	+5.34%
TRAIN(<oos)	1896	52		+1.18	+1.31	+0.13		-38.1%	+1.7%		-0.58	60	+21.62%
OOS	617	16		+1.42	+1.50	+0.08		-39.5%	-0.6%		-0.00	50	+5.22%

YEARLY WALK-FORWARD

year	days	on		bookSh	hedSh	dSh		bookDD	dDD		onBook%	paid%	sleeve%
2016	136	1		+2.73	+2.72	-0.01		-6.0%	+0.0%		-0.17	0	-0.13%
2017	251	0		+1.82	+1.82	+0.00		-10.8%	+0.0%		na	na	+0.00%
2018	251	8		+0.71	+0.80	+0.09		-28.9%	-0.1%		-0.69	62	+2.14%
2019	252	4		+2.53	+2.64	+0.10		-10.1%	+1.6%		-0.80	75	+1.47%
2020	253	11		+1.44	+1.99	+0.56		-33.8%	+10.8%		-2.03	73	+17.21%
2021	252	7		+0.49	+0.58	+0.09		-21.6%	+1.7%		-0.41	71	+2.41%
2022	251	16		-0.76	-0.82	-0.06		-31.8%	+0.4%		0.15	50	-1.05%
2023	250	5		+2.23	+2.22	-0.01		-16.8%	-0.2%		0.33	40	-0.43%
2024	252	6		+2.12	+2.20	+0.08		-20.4%	+2.1%		0.60	67	+1.90%
2025	250	6		-0.15	-0.04	+0.11		-37.1%	-0.6%		-0.44	50	+3.58%
2026	115	4		+3.01	+3.03	+0.02		-12.1%	+0.0%		-0.24	25	-0.25%

EXAMPLE CASE STUDY - 2026-06-16

Trigger @ 2026-06-15 close: QQQ +3.1% up-shock, SPY 5d-vol $z = +1.46$ (elevated) -> RULE FIRES

Next day 2026-06-16: book -3.80% (leaders-led: top10 -5.7%, SMH -4.8%, QQQ -1.9%, SPY -0.6%; spread -4.0%)

Hedge result: SOXS (inv-vol, 12% of exposure = ~\$12k) -> +2.03% (\$+2,034 on \$100k) -> the -3.8% book day softened to ~ -1.8%

Appendix A

Correlation & Gap Analysis (full history 2016-2026)

(source: corr_tails_by_horizon.pdf)

Trailing-factor avg, correlation & UP-DN gap, by forward book-move segment

UP/DN cells: top = mean of trailing measure in-segment; bottom = its corr with book fwd return (purple shade = |corr|).

D = UP avg - DN avg (pp for %-measures; raw for scores). D cells shaded amber by |D| within each column (deepest = biggest gap). See next page for sorted ranking.

	FWD 1D			FWD 5D			FWD 20D		
	thr +/-2.0%			thr +/-4.5%			thr +/-7.5%		
TRAILING MEASURE	UP n=296	DN n=250	D(U-D)	UP n=340	DN n=222	D(U-D)	UP n=655	DN n=215	D(U-D)
BOOK									
book_1d	-0.2% r=-0.05	+0.0% r=-0.03	-0.3	-0.1% r=-0.01	+0.2% r=0.12	-0.2	+0.2% r=-0.02	+0.1% r=0.02	+0.1
book_5d	-0.1% r=-0.03	+0.2% r=0.16	-0.3	-0.1% r=-0.02	+0.1% r=0.02	-0.2	+0.9% r=0.04	+0.4% r=-0.09	+0.5
book_20d	+2.8% r=-0.01	+2.2% r=0.14	+0.6	+2.8% r=0.12	+1.5% r=0.06	+1.3	+3.0% r=-0.03	+1.2% r=-0.08	+1.8
LONG/SHORT SPREAD									
spread_1d	-0.2% r=0.04	-0.0% r=-0.09	-0.1	-0.1% r=-0.03	+0.1% r=-0.11	-0.2	-0.0% r=-0.03	+0.2% r=-0.01	-0.2
spread_5d	-0.1% r=0.09	+0.3% r=-0.16	-0.4	-0.2% r=-0.15	-0.1% r=-0.29	-0.0	-0.1% r=-0.01	-0.3% r=-0.22	+0.2
spread_20d	+0.2% r=0.10	+0.6% r=-0.05	-0.5	+0.4% r=-0.01	+0.1% r=-0.13	+0.3	-0.8% r=0.11	+0.1% r=-0.08	-0.9
BASKETS									
top10_1d	-0.3% r=-0.06	-0.0% r=-0.03	-0.3	-0.0% r=-0.02	+0.1% r=0.07	-0.2	+0.2% r=-0.05	+0.2% r=0.01	-0.0
top10_5d	-0.2% r=0.03	+0.2% r=0.13	-0.4	-0.0% r=-0.04	+0.2% r=-0.03	-0.3	+0.9% r=0.01	+0.6% r=-0.13	+0.3
top10_20d	+2.3% r=0.01	+2.2% r=0.13	+0.1	+2.8% r=0.09	+1.7% r=0.03	+1.0	+3.0% r=-0.05	+1.7% r=-0.02	+1.3
bottom10_1d	-0.1% r=-0.09	+0.0% r=0.05	-0.1	+0.0% r=0.01	+0.0% r=0.14	+0.0	+0.2% r=-0.01	+0.1% r=0.01	+0.2
bottom10_5d	-0.0% r=-0.04	-0.0% r=0.24	+0.0	+0.1% r=0.09	+0.3% r=0.22	-0.2	+1.0% r=0.01	+0.9% r=0.06	+0.1
bottom10_20d	+2.1% r=-0.07	+1.5% r=0.17	+0.6	+2.3% r=0.10	+1.6% r=0.15	+0.8	+3.8% r=-0.14	+1.6% r=0.05	+2.2
universe_1d	-0.2% r=-0.08	-0.0% r=0.00	-0.2	-0.0% r=0.03	+0.1% r=0.12	-0.1	+0.2% r=-0.03	+0.1% r=0.04	+0.1
universe_5d	-0.1% r=-0.01	+0.0% r=0.21	-0.1	-0.1% r=0.08	+0.2% r=0.16	-0.3	+0.8% r=0.01	+0.7% r=0.03	+0.1
universe_20d	+1.6% r=-0.06	+1.3% r=0.18	+0.3	+1.9% r=0.09	+1.3% r=0.13	+0.6	+2.8% r=-0.14	+1.8% r=0.01	+1.0
ETFs									
spy_1d	-0.2% r=-0.11	+0.0% r=-0.03	-0.3	-0.1% r=0.02	+0.1% r=0.10	-0.2	+0.1% r=-0.04	+0.0% r=0.03	+0.1
spy_5d	-0.2% r=-0.08	-0.1% r=0.25	-0.1	-0.2% r=0.03	+0.0% r=0.24	-0.2	+0.4% r=0.02	+0.2% r=0.06	+0.2
spy_20d	+0.3% r=-0.13	+0.2% r=0.24	+0.1	+0.5% r=0.09	+0.3% r=0.29	+0.1	+1.2% r=-0.12	+0.8% r=0.10	+0.5
qqq_1d	-0.3% r=-0.11	+0.0% r=-0.02	-0.3	-0.1% r=0.00	+0.1% r=0.09	-0.2	+0.1% r=-0.04	+0.1% r=0.04	+0.0
qqq_5d	-0.2% r=-0.05	+0.0% r=0.20	-0.2	-0.2% r=0.05	+0.1% r=0.18	-0.3	+0.4% r=0.04	+0.3% r=0.00	+0.1
qqq_20d	+0.6% r=-0.09	+0.6% r=0.18	+0.0	+0.8% r=0.12	+0.8% r=0.19	-0.1	+1.6% r=-0.10	+1.1% r=0.02	+0.5
smh_1d	-0.3% r=-0.03	+0.1% r=-0.01	-0.4	+0.1% r=-0.00	+0.2% r=0.07	-0.2	+0.2% r=-0.02	+0.1% r=0.01	+0.1
smh_5d	-0.0% r=0.03	+0.3% r=0.18	-0.3	-0.0% r=0.04	+0.3% r=0.17	-0.3	+0.7% r=0.06	+0.5% r=-0.01	+0.3
smh_20d	+2.2% r=-0.01	+1.8% r=0.15	+0.3	+1.8% r=0.15	+1.4% r=0.20	+0.4	+2.5% r=-0.02	+1.1% r=0.20	+1.3
psq_1d	+0.3% r=0.10	-0.0% r=0.03	+0.3	+0.1% r=-0.00	-0.1% r=-0.09	+0.2	-0.1% r=0.04	-0.0% r=-0.04	-0.0
psq_5d	+0.3% r=0.05	+0.0% r=-0.18	+0.3	+0.2% r=-0.04	-0.1% r=-0.16	+0.4	-0.4% r=-0.04	-0.2% r=-0.01	-0.1
psq_20d	-0.4% r=0.09	-0.4% r=-0.19	+0.0	-0.6% r=-0.13	-0.6% r=-0.20	+0.1	-1.4% r=0.12	-0.8% r=-0.01	-0.7
soxs_1d	+1.1% r=0.03	-0.2% r=0.02	+1.3	-0.2% r=0.01	-0.7% r=-0.05	+0.6	-0.6% r=0.02	-0.3% r=-0.01	-0.3
soxs_5d	+0.1% r=-0.04	-1.5% r=-0.15	+1.6	-0.0% r=-0.01	-1.4% r=-0.12	+1.4	-2.3% r=-0.07	-1.5% r=-0.01	-0.8
soxs_20d	-7.4% r=0.02	-6.8% r=-0.16	-0.5	-6.1% r=-0.12	-5.9% r=-0.20	-0.2	-8.3% r=0.03	-5.1% r=-0.18	-3.2
AVG SCORE									
sc.top10_1d	0.835 r=0.08	0.835 r=0.01	+0.001	0.836 r=-0.14	0.832 r=-0.12	+0.004	0.836 r=0.15	0.831 r=-0.12	+0.005
sc.top10_5d	0.836 r=0.07	0.834 r=0.04	+0.002	0.836 r=0.10	0.832 r=-0.03	+0.004	0.835 r=0.13	0.830 r=-0.11	+0.005
sc.top10_20d	0.837 r=0.08	0.836 r=0.04	+0.001	0.837 r=0.16	0.835 r=0.02	+0.002	0.834 r=0.16	0.835 r=-0.14	-0.001
sc.bottom10_1d	0.209 r=-0.06	0.208 r=0.02	+0.000	0.212 r=-0.11	0.212 r=0.20	-0.001	0.210 r=-0.02	0.210 r=0.11	-0.000
sc.bottom10_5d	0.209 r=-0.05	0.207 r=-0.02	+0.001	0.211 r=-0.05	0.210 r=0.14	+0.001	0.210 r=-0.00	0.208 r=0.01	+0.002
sc.bottom10_20d	0.209 r=-0.01	0.208 r=0.00	+0.000	0.209 r=0.05	0.208 r=-0.02	+0.001	0.210 r=-0.06	0.207 r=0.08	+0.003
sc.universe_1d	0.507 r=-0.07	0.507 r=0.02	-0.000	0.507 r=-0.09	0.506 r=-0.06	+0.000	0.507 r=-0.31	0.506 r=-0.13	+0.000
sc.universe_5d	0.507 r=-0.07	0.507 r=0.01	-0.000	0.507 r=-0.09	0.506 r=-0.06	+0.000	0.507 r=-0.31	0.506 r=-0.13	+0.000
sc.universe_20d	0.506 r=-0.05	0.507 r=-0.00	-0.000	0.507 r=-0.08	0.506 r=-0.06	+0.000	0.507 r=-0.30	0.506 r=-0.12	+0.000
sc.held_1d	0.613 r=0.11	0.615 r=-0.06	-0.001	0.612 r=0.03	0.610 r=-0.34	+0.002	0.604 r=0.13	0.606 r=-0.28	-0.002
sc.held_5d	0.615 r=0.09	0.613 r=-0.07	+0.001	0.613 r=0.06	0.609 r=-0.24	+0.004	0.602 r=0.14	0.605 r=-0.21	-0.002
sc.held_20d	0.613 r=0.13	0.612 r=-0.05	+0.001	0.608 r=0.18	0.608 r=-0.20	+0.001	0.594 r=0.16	0.612 r=-0.05	-0.018
REALIZED VOL (ann.)									
spy_vol_1d	+14.1% r=0.13	+14.5% r=-0.22	-0.4	+14.0% r=0.00	+15.3% r=-0.32	-1.3	+13.4% r=0.08	+13.6% r=-0.04	-0.1
spy_vol_5d	+16.7% r=0.20	+17.7% r=-0.26	-1.0	+16.9% r=0.00	+17.0% r=-0.36	-0.1	+16.5% r=0.12	+15.1% r=-0.05	+1.4
spy_vol_20d	+17.8% r=0.16	+16.8% r=-0.17	+1.0	+18.4% r=0.02	+15.5% r=-0.30	+2.9	+17.5% r=0.11	+15.3% r=-0.00	+2.2
qqq_vol_1d	+18.3% r=0.10	+18.7% r=-0.17	-0.4	+18.1% r=-0.02	+19.4% r=-0.25	-1.3	+16.5% r=0.09	+18.5% r=0.02	-2.1
qqq_vol_5d	+21.6% r=0.17	+22.9% r=-0.22	-1.3	+21.8% r=-0.00	+22.1% r=-0.33	-0.3	+20.3% r=0.14	+21.1% r=0.02	-0.8
qqq_vol_20d	+22.8% r=0.14	+22.0% r=-0.15	+0.8	+23.2% r=0.02	+21.0% r=-0.27	+2.2	+21.6% r=0.12	+21.4% r=0.06	+0.2

Read across a horizon group: UP, DN, then gap D. Positive D = factor ran higher before up-moves. In-segment corr is range-compressed by tail conditioning.

Biggest UP-DN trailing-factor gaps, sorted by |D| within each horizon

D = UP avg - DN avg (pp for %-measures, raw for scores). Each column sorted by absolute gap, largest first. Scores sit on a 0-1 scale, so their raw gaps look tiny next to % returns.

FWD 1D (thr +/-2.0%)

measure	D	(U avg / D avg)
1.soxs_5d	+1.6	+0.1%/ -1.5%
2.soxs_1d	+1.3	+1.1%/ -0.2%
3.qqq_vol_5d	-1.3	+21.6%/+22.9%
4.spy_vol_5d	-1.0	+16.7%/+17.7%
5.spy_vol_20d	+1.0	+17.8%/+16.8%
6.qqq_vol_20d	+0.8	+22.8%/+22.0%
7.book_20d	+0.6	+2.8%/ +2.2%
8.bottom10_20d	+0.6	+2.1%/ +1.5%
9.soxs_20d	-0.5	-7.4%/ -6.8%
10.spread_20d	-0.5	+0.2%/ +0.6%
11.spy_vol_1d	-0.4	+14.1%/+14.5%
12.spread_5d	-0.4	-0.1%/ +0.3%
13.qqq_vol_1d	-0.4	+18.3%/+18.7%
14.top10_5d	-0.4	-0.2%/ +0.2%
15.smh_1d	-0.4	-0.3%/ +0.1%
16.smh_20d	+0.3	+2.2%/ +1.8%
17.book_5d	-0.3	-0.1%/ +0.2%
18.smh_5d	-0.3	-0.0%/ +0.3%
19.universe_20d	+0.3	+1.6%/ +1.3%
20.psq_5d	+0.3	+0.3%/ +0.0%
21.psq_1d	+0.3	+0.3%/ -0.0%
22.qqq_1d	-0.3	-0.3%/ +0.0%
23.top10_1d	-0.3	-0.3%/ -0.0%
24.spy_1d	-0.3	-0.2%/ +0.0%
25.book_1d	-0.3	-0.2%/ +0.0%
26.qqq_5d	-0.2	-0.2%/ +0.0%
27.universe_1d	-0.2	-0.2%/ -0.0%
28.sc.top10_5d	+0.2	+83.6%/+83.4%
29.sc.top10_20d	+0.1	+83.7%/+83.6%
30.spy_5d	-0.1	-0.2%/ -0.1%
31.sc.bottom10_5d	+0.1	+20.9%/+20.7%
32.bottom10_1d	-0.1	-0.1%/ +0.0%
33.spread_1d	-0.1	-0.2%/ -0.0%
34.sc.held_1d	-0.1	+61.3%/+61.5%
35.universe_5d	-0.1	-0.1%/ +0.0%
36.top10_20d	+0.1	+2.3%/ +2.2%
37.sc.held_5d	+0.1	+61.5%/+61.3%
38.sc.held_20d	+0.1	+61.3%/+61.2%
39.spy_20d	+0.1	+0.3%/ +0.2%
40.sc.top10_1d	+0.1	+83.5%/+83.5%
41.sc.bottom10_20d	+0.0	+20.9%/+20.8%
42.sc.bottom10_1d	+0.0	+20.9%/+20.8%
43.psq_20d	+0.0	-0.4%/ -0.4%
44.bottom10_5d	+0.0	-0.0%/ -0.0%
45.sc.universe_20d	-0.0	+50.6%/+50.7%
46.sc.universe_1d	-0.0	+50.7%/+50.7%
47.sc.universe_5d	-0.0	+50.7%/+50.7%
48.qqq_20d	+0.0	+0.6%/ +0.6%

FWD 5D (thr +/-4.5%)

measure	D	(U avg / D avg)
1.spy_vol_20d	+2.9	+18.4%/+15.5%
2.qqq_vol_20d	+2.2	+23.2%/+21.0%
3.soxs_5d	+1.4	-0.0%/ -1.4%
4.book_20d	+1.3	+2.8%/ +1.5%
5.spy_vol_1d	-1.3	+14.0%/+15.3%
6.qqq_vol_1d	-1.3	+18.1%/+19.4%
7.top10_20d	+1.0	+2.8%/ +1.7%
8.bottom10_20d	+0.8	+2.3%/ +1.6%
9.universe_20d	+0.6	+1.9%/ +1.3%
10.soxs_1d	+0.6	-0.2%/ -0.7%
11.sc.held_5d	+0.4	+61.3%/+60.9%
12.sc.top10_1d	+0.4	+83.6%/+83.2%
13.smh_20d	+0.4	+1.8%/ +1.4%
14.sc.top10_5d	+0.4	+83.6%/+83.2%
15.psq_5d	+0.4	+0.2%/ -0.1%
16.qqq_5d	-0.3	-0.2%/ +0.1%
17.spread_20d	+0.3	+0.4%/ +0.1%
18.qqq_vol_5d	-0.3	+21.8%/+22.1%
19.universe_5d	-0.3	-0.1%/ +0.2%
20.smh_5d	-0.3	-0.0%/ +0.3%
21.top10_5d	-0.3	-0.0%/ +0.2%
22.book_1d	-0.2	-0.1%/ +0.2%
23.psq_1d	+0.2	+0.1%/ -0.1%
24.qqq_1d	-0.2	-0.1%/ +0.1%
25.bottom10_5d	-0.2	+0.1%/ +0.3%
26.book_5d	-0.2	-0.1%/ +0.1%
27.sc.top10_20d	+0.2	+83.7%/+83.5%
28.soxs_20d	-0.2	-6.1%/ -5.9%
29.spy_5d	-0.2	-0.2%/ +0.0%
30.spread_1d	-0.2	-0.1%/ +0.1%
31.smh_1d	-0.2	+0.1%/ +0.2%
32.spy_1d	-0.2	-0.1%/ +0.1%
33.top10_1d	-0.2	-0.0%/ +0.1%
34.sc.held_1d	+0.2	+61.2%/+61.0%
35.spy_20d	+0.1	+0.5%/ +0.3%
36.universe_1d	-0.1	-0.0%/ +0.1%
37.qqq_20d	-0.1	+0.8%/ +0.8%
38.sc.bottom10_20d	+0.1	+20.9%/+20.8%
39.psq_20d	+0.1	-0.6%/ -0.6%
40.sc.bottom10_1d	-0.1	+21.2%/+21.2%
41.sc.held_20d	+0.1	+60.8%/+60.8%
42.sc.bottom10_5d	+0.1	+21.1%/+21.0%
43.spy_vol_5d	-0.1	+16.9%/+17.0%
44.spread_5d	-0.0	-0.2%/ -0.1%
45.bottom10_1d	+0.0	+0.0%/ +0.0%
46.sc.universe_5d	+0.0	+50.7%/+50.6%
47.sc.universe_1d	+0.0	+50.7%/+50.6%
48.sc.universe_20d	+0.0	+50.7%/+50.6%

FWD 20D (thr +/-7.5%)

measure	D	(U avg / D avg)
1.soxs_20d	-3.2	-8.3%/ -5.1%
2.bottom10_20d	+2.2	+3.8%/ +1.6%
3.spy_vol_20d	+2.2	+17.5%/+15.3%
4.qqq_vol_1d	-2.1	+16.5%/+18.5%
5.book_20d	+1.8	+3.0%/ +1.2%
6.sc.held_20d	-1.8	+59.4%/+61.2%
7.spy_vol_5d	+1.4	+16.5%/+15.1%
8.smh_20d	+1.3	+2.5%/ +1.1%
9.top10_20d	+1.3	+3.0%/ +1.7%
10.universe_20d	+1.0	+2.8%/ +1.8%
11.spread_20d	-0.9	-0.8%/ +0.1%
12.qqq_vol_5d	-0.8	+20.3%/+21.1%
13.soxs_5d	-0.8	-2.3%/ -1.5%
14.psq_20d	-0.7	-1.4%/ -0.8%
15.qqq_20d	+0.5	+1.6%/ +1.1%
16.book_5d	+0.5	+0.9%/ +0.4%
17.sc.top10_5d	+0.5	+83.5%/+83.0%
18.sc.top10_1d	+0.5	+83.6%/+83.1%
19.spy_20d	+0.5	+1.2%/ +0.8%
20.soxs_1d	-0.3	-0.6%/ -0.3%
21.sc.bottom10_20d	+0.3	+21.0%/+20.7%
22.top10_5d	+0.3	+0.9%/ +0.6%
23.smh_5d	+0.3	+0.7%/ +0.5%
24.sc.held_1d	-0.2	+60.4%/+60.6%
25.sc.held_5d	-0.2	+60.2%/+60.5%
26.qqq_vol_20d	+0.2	+21.6%/+21.4%
27.spread_1d	-0.2	-0.0%/ +0.2%
28.spy_5d	+0.2	+0.4%/ +0.2%
29.sc.bottom10_5d	+0.2	+21.0%/+20.8%
30.bottom10_1d	+0.2	+0.2%/ +0.1%
31.spread_5d	+0.2	-0.1%/ -0.3%
32.spy_vol_1d	-0.1	+13.4%/+13.6%
33.psq_5d	-0.1	-0.4%/ -0.2%
34.bottom10_5d	+0.1	+1.0%/ +0.9%
35.sc.top10_20d	-0.1	+83.4%/+83.5%
36.smh_1d	+0.1	+0.2%/ +0.1%
37.qqq_5d	+0.1	+0.4%/ +0.3%
38.book_1d	+0.1	+0.2%/ +0.1%
39.universe_5d	+0.1	+0.8%/ +0.7%
40.spy_1d	+0.1	+0.1%/ +0.0%
41.universe_1d	+0.1	+0.2%/ +0.1%
42.psq_1d	-0.0	-0.1%/ -0.0%
43.qqq_1d	+0.0	+0.1%/ +0.1%
44.sc.universe_5d	+0.0	+50.7%/+50.6%
45.sc.universe_1d	+0.0	+50.7%/+50.6%
46.sc.universe_20d	+0.0	+50.7%/+50.6%
47.top10_1d	-0.0	+0.2%/ +0.2%
48.sc.bottom10_1d	-0.0	+21.0%/+21.0%

Trailing -> Forward correlation call-outs (|r| >= 0.5)

Strong = |r| >= 0.70 | Moderate = 0.50 <= |r| < 0.70. Cells ranked by |r|.

avg() = mean of that factor across the days passing the panel filter.

Small-n panels (n < ~40) are noise-dominated; weight accordingly.

Forward 1-Day portfolio return (filtered by book_fwd_1d)

UP > +2.0% (n=296)

MODERATE (0.50-0.70):

-0.62	spy_vol_trl_5d	avg=+16.7%	->	psq_fwd_1d	avg=-1.7%
-0.61	qqq_vol_trl_5d	avg=+21.6%	->	psq_fwd_1d	avg=-1.7%
-0.58	qqq_vol_trl_20d	avg=+22.8%	->	psq_fwd_1d	avg=-1.7%
+0.58	spy_trl_20d	avg=+0.3%	->	psq_fwd_1d	avg=-1.7%
+0.57	qqq_vol_trl_20d	avg=+22.8%	->	universe_fwd_1d	avg=+2.3%
-0.57	spy_vol_trl_20d	avg=+17.8%	->	psq_fwd_1d	avg=-1.7%
+0.55	qqq_vol_trl_5d	avg=+21.6%	->	universe_fwd_1d	avg=+2.3%
+0.55	spy_vol_trl_5d	avg=+16.7%	->	universe_fwd_1d	avg=+2.3%
+0.54	spy_vol_trl_20d	avg=+17.8%	->	universe_fwd_1d	avg=+2.3%
+0.54	qqq_trl_20d	avg=+0.6%	->	psq_fwd_1d	avg=-1.7%
-0.54	spy_trl_20d	avg=+0.3%	->	universe_fwd_1d	avg=+2.3%
-0.54	universe_trl_20d	avg=+1.6%	->	bottom10_fwd_1d	avg=+2.4%
+0.53	universe_trl_20d	avg=+1.6%	->	psq_fwd_1d	avg=-1.7%
-0.53	psq_trl_20d	avg=-0.4%	->	psq_fwd_1d	avg=-1.7%
-0.53	qqq_trl_20d	avg=+0.6%	->	bottom10_fwd_1d	avg=+2.4%
-0.53	book_trl_20d	avg=+2.8%	->	bottom10_fwd_1d	avg=+2.4%
-0.53	qqq_trl_20d	avg=+0.6%	->	universe_fwd_1d	avg=+2.3%
-0.53	universe_trl_20d	avg=+1.6%	->	universe_fwd_1d	avg=+2.3%
-0.52	qqq_vol_trl_5d	avg=+21.6%	->	soxs_fwd_1d	avg=-7.7%
+0.52	psq_trl_20d	avg=-0.4%	->	universe_fwd_1d	avg=+2.3%
+0.52	book_trl_20d	avg=+2.8%	->	psq_fwd_1d	avg=-1.7%
+0.51	psq_trl_20d	avg=-0.4%	->	bottom10_fwd_1d	avg=+2.4%
+0.51	qqq_vol_trl_20d	avg=+22.8%	->	bottom10_fwd_1d	avg=+2.4%
-0.51	spy_vol_trl_5d	avg=+16.7%	->	soxs_fwd_1d	avg=-7.7%
-0.50	book_trl_20d	avg=+2.8%	->	universe_fwd_1d	avg=+2.3%
+0.50	qqq_vol_trl_5d	avg=+21.6%	->	smh_fwd_1d	avg=+2.5%

UP > +3.5% (n=68)

MODERATE (0.50-0.70):

-0.70	book_trl_20d	avg=+2.5%	->	bottom10_fwd_1d	avg=+3.1%
-0.67	qqq_trl_20d	avg=-0.3%	->	bottom10_fwd_1d	avg=+3.1%
-0.67	qqq_vol_trl_20d	avg=+25.1%	->	psq_fwd_1d	avg=-2.6%
+0.66	book_trl_20d	avg=+2.5%	->	spread_fwd_1d	avg=+0.5%
-0.66	universe_trl_20d	avg=+0.7%	->	bottom10_fwd_1d	avg=+3.1%
+0.66	spy_trl_20d	avg=-1.1%	->	psq_fwd_1d	avg=-2.6%
-0.66	spy_trl_20d	avg=-1.1%	->	universe_fwd_1d	avg=+3.1%
-0.65	qqq_trl_20d	avg=-0.3%	->	universe_fwd_1d	avg=+3.1%
-0.65	book_trl_20d	avg=+2.5%	->	universe_fwd_1d	avg=+3.1%
-0.65	universe_trl_20d	avg=+0.7%	->	universe_fwd_1d	avg=+3.1%
+0.65	spy_trl_20d	avg=-1.1%	->	spread_fwd_20d	avg=-1.3%
-0.64	spy_vol_trl_20d	avg=+20.6%	->	psq_fwd_1d	avg=-2.6%
+0.64	psq_trl_20d	avg=+0.6%	->	bottom10_fwd_1d	avg=+3.1%
-0.64	bottom10_trl_20d	avg=+0.7%	->	bottom10_fwd_1d	avg=+3.1%
+0.64	psq_trl_20d	avg=+0.6%	->	universe_fwd_1d	avg=+3.1%
-0.63	spy_trl_20d	avg=-1.1%	->	bottom10_fwd_1d	avg=+3.1%
-0.63	bottom10_trl_20d	avg=+0.7%	->	universe_fwd_1d	avg=+3.1%
+0.63	universe_trl_20d	avg=+0.7%	->	psq_fwd_1d	avg=-2.6%
+0.63	book_trl_20d	avg=+2.5%	->	spread_fwd_20d	avg=-1.3%
+0.63	qqq_trl_20d	avg=-0.3%	->	psq_fwd_1d	avg=-2.6%
+0.63	qqq_vol_trl_20d	avg=+25.1%	->	universe_fwd_1d	avg=+3.1%
-0.62	spy_vol_trl_5d	avg=+21.8%	->	psq_fwd_1d	avg=-2.6%

+0.62	qqq_vol_trl_20d	avg=+25.1%	->	bottom10_fwd_1d	avg=+3.1%
+0.62	book_trl_20d	avg=+2.5%	->	psq_fwd_1d	avg=-2.6%
-0.61	qqq_vol_trl_5d	avg=+26.4%	->	psq_fwd_1d	avg=-2.6%
+0.61	spy_vol_trl_20d	avg=+20.6%	->	universe_fwd_1d	avg=+3.1%
-0.61	psq_trl_20d	avg=+0.6%	->	psq_fwd_1d	avg=-2.6%
+0.60	qqq_trl_20d	avg=-0.3%	->	spread_fwd_20d	avg=-1.3%
-0.60	qqq_vol_trl_5d	avg=+26.4%	->	spread_fwd_20d	avg=-1.3%
+0.60	spy_vol_trl_20d	avg=+20.6%	->	bottom10_fwd_1d	avg=+3.1%
+0.60	qqq_vol_trl_20d	avg=+25.1%	->	bottom10_fwd_20d	avg=+8.2%
-0.60	top10_trl_20d	avg=+2.4%	->	bottom10_fwd_1d	avg=+3.1%
-0.60	spy_vol_trl_5d	avg=+21.8%	->	spread_fwd_20d	avg=-1.3%
+0.59	spy_vol_trl_20d	avg=+20.6%	->	bottom10_fwd_20d	avg=+8.2%
-0.58	smh_trl_20d	avg=+1.8%	->	bottom10_fwd_1d	avg=+3.1%
+0.58	bottom10_trl_20d	avg=+0.7%	->	psq_fwd_1d	avg=-2.6%
-0.58	psq_trl_20d	avg=+0.6%	->	spread_fwd_20d	avg=-1.3%
+0.58	qqq_trl_20d	avg=-0.3%	->	spread_fwd_1d	avg=+0.5%
+0.57	spy_trl_5d	avg=-0.7%	->	psq_fwd_1d	avg=-2.6%
-0.57	qqq_vol_trl_20d	avg=+25.1%	->	spread_fwd_20d	avg=-1.3%
+0.57	universe_trl_20d	avg=+0.7%	->	spread_fwd_1d	avg=+0.5%
+0.57	spy_vol_trl_5d	avg=+21.8%	->	universe_fwd_1d	avg=+3.1%
-0.57	smh_trl_20d	avg=+1.8%	->	universe_fwd_1d	avg=+3.1%
-0.57	top10_trl_20d	avg=+2.4%	->	universe_fwd_1d	avg=+3.1%
+0.56	bottom10_trl_20d	avg=+0.7%	->	spread_fwd_20d	avg=-1.3%
-0.56	spy_vol_trl_20d	avg=+20.6%	->	spread_fwd_20d	avg=-1.3%
+0.56	qqq_vol_trl_5d	avg=+26.4%	->	universe_fwd_1d	avg=+3.1%
+0.56	book_trl_5d	avg=-0.1%	->	spread_fwd_1d	avg=+0.5%
+0.55	smh_trl_20d	avg=+1.8%	->	spread_fwd_20d	avg=-1.3%
+0.55	universe_trl_20d	avg=+0.7%	->	spread_fwd_20d	avg=-1.3%
+0.55	bottom10_trl_20d	avg=+0.7%	->	spread_fwd_1d	avg=+0.5%
-0.54	qqq_vol_trl_20d	avg=+25.1%	->	psq_fwd_20d	avg=-4.7%
+0.54	qqq_vol_trl_20d	avg=+25.1%	->	smh_fwd_1d	avg=+3.7%
-0.54	qqq_vol_trl_20d	avg=+25.1%	->	soxs_fwd_1d	avg=-11.6%
-0.54	spy_vol_trl_20d	avg=+20.6%	->	psq_fwd_20d	avg=-4.7%
-0.54	spy_vol_trl_1d	avg=+19.4%	->	spread_fwd_20d	avg=-1.3%
-0.54	psq_trl_20d	avg=+0.6%	->	spread_fwd_1d	avg=+0.5%
+0.54	book_trl_5d	avg=-0.1%	->	psq_fwd_1d	avg=-2.6%
-0.53	qqq_vol_trl_5d	avg=+26.4%	->	soxs_fwd_1d	avg=-11.6%
+0.53	smh_trl_20d	avg=+1.8%	->	psq_fwd_1d	avg=-2.6%
+0.53	top10_trl_20d	avg=+2.4%	->	psq_fwd_1d	avg=-2.6%
+0.52	spy_vol_trl_20d	avg=+20.6%	->	smh_fwd_1d	avg=+3.7%
-0.52	spy_trl_20d	avg=-1.1%	->	smh_fwd_1d	avg=+3.7%
+0.52	top10_trl_20d	avg=+2.4%	->	spread_fwd_1d	avg=+0.5%
-0.52	spy_vol_trl_5d	avg=+21.8%	->	soxs_fwd_1d	avg=-11.6%
+0.52	spy_trl_20d	avg=-1.1%	->	soxs_fwd_1d	avg=-11.6%
+0.52	smh_trl_20d	avg=+1.8%	->	spread_fwd_1d	avg=+0.5%
-0.52	spy_vol_trl_20d	avg=+20.6%	->	soxs_fwd_1d	avg=-11.6%
-0.52	spy_trl_5d	avg=-0.7%	->	universe_fwd_1d	avg=+3.1%
-0.52	score_top10_trl_20d	avg=0.840	->	universe_fwd_1d	avg=+3.1%
+0.52	spy_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_1d	avg=+3.1%
+0.52	qqq_vol_trl_5d	avg=+26.4%	->	bottom10_fwd_20d	avg=+8.2%
+0.52	spy_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_20d	avg=+8.2%
+0.51	qqq_trl_5d	avg=-0.6%	->	psq_fwd_1d	avg=-2.6%
-0.51	psq_trl_5d	avg=+0.7%	->	psq_fwd_1d	avg=-2.6%
-0.51	book_trl_5d	avg=-0.1%	->	bottom10_fwd_1d	avg=+3.1%
+0.51	qqq_vol_trl_5d	avg=+26.4%	->	bottom10_fwd_1d	avg=+3.1%
+0.50	spy_trl_20d	avg=-1.1%	->	spread_fwd_1d	avg=+0.5%
+0.50	spy_vol_trl_5d	avg=+21.8%	->	smh_fwd_1d	avg=+3.7%
+0.50	qqq_vol_trl_5d	avg=+26.4%	->	smh_fwd_1d	avg=+3.7%

-0.50	score_top10_trl_20d	avg=0.840	->	bottom10_fwd_1d	avg=+3.1%
-0.50	universe_trl_20d	avg=+0.7%	->	smh_fwd_1d	avg=+3.7%

DOWN < -2.0% (n=250)

MODERATE (0.50-0.70):

+0.53	spy_vol_trl_5d	avg=+17.7%	->	psq_fwd_1d	avg=+2.1%
+0.51	top10_trl_20d	avg=+2.2%	->	bottom10_fwd_1d	avg=-2.7%
-0.51	spy_vol_trl_5d	avg=+17.7%	->	universe_fwd_1d	avg=-2.7%
+0.51	qqq_vol_trl_5d	avg=+22.9%	->	psq_fwd_1d	avg=+2.1%
+0.51	book_trl_20d	avg=+2.2%	->	bottom10_fwd_1d	avg=-2.7%

DOWN < -3.5% (n=77)

MODERATE (0.50-0.70):

+0.63	spy_vol_trl_5d	avg=+21.9%	->	psq_fwd_1d	avg=+3.1%
+0.62	spy_vol_trl_1d	avg=+17.4%	->	psq_fwd_1d	avg=+3.1%
-0.62	qqq_vol_trl_5d	avg=+26.8%	->	spread_fwd_20d	avg=-2.9%
+0.60	qqq_vol_trl_5d	avg=+26.8%	->	psq_fwd_1d	avg=+3.1%
-0.59	spy_vol_trl_5d	avg=+21.9%	->	spread_fwd_20d	avg=-2.9%
-0.58	spy_vol_trl_1d	avg=+17.4%	->	universe_fwd_1d	avg=-3.8%
-0.58	qqq_vol_trl_20d	avg=+24.0%	->	spread_fwd_20d	avg=-2.9%
+0.57	qqq_vol_trl_1d	avg=+21.2%	->	psq_fwd_1d	avg=+3.1%
-0.54	spy_vol_trl_20d	avg=+19.1%	->	spread_fwd_20d	avg=-2.9%
+0.53	spy_vol_trl_20d	avg=+19.1%	->	psq_fwd_1d	avg=+3.1%
-0.53	spy_vol_trl_5d	avg=+21.9%	->	universe_fwd_1d	avg=-3.8%
-0.53	qqq_vol_trl_1d	avg=+21.2%	->	universe_fwd_1d	avg=-3.8%
+0.52	spy_vol_trl_5d	avg=+21.9%	->	bottom10_fwd_20d	avg=+1.4%
+0.52	spy_vol_trl_1d	avg=+17.4%	->	soxs_fwd_1d	avg=+13.7%
+0.52	qqq_vol_trl_20d	avg=+24.0%	->	psq_fwd_1d	avg=+3.1%
+0.52	spy_trl_20d	avg=-1.0%	->	spread_fwd_20d	avg=-2.9%
-0.52	spy_vol_trl_1d	avg=+17.4%	->	smh_fwd_1d	avg=-4.4%
+0.51	spy_vol_trl_20d	avg=+19.1%	->	bottom10_fwd_20d	avg=+1.4%
-0.50	smh_trl_20d	avg=+0.9%	->	psq_fwd_1d	avg=+3.1%
+0.50	book_trl_20d	avg=+1.1%	->	bottom10_fwd_1d	avg=-4.0%
+0.50	qqq_vol_trl_5d	avg=+26.8%	->	bottom10_fwd_20d	avg=+1.4%

Forward 5-Day portfolio return (filtered by book_fwd_5d)

UP > +4.5% (n=340)

MODERATE (0.50-0.70):

-0.58	qqq_vol_trl_20d	avg=+23.2%	->	psq_fwd_5d	avg=-3.4%
+0.58	qqq_vol_trl_20d	avg=+23.2%	->	bottom10_fwd_5d	avg=+5.1%
+0.57	qqq_vol_trl_20d	avg=+23.2%	->	universe_fwd_5d	avg=+5.0%
-0.56	spy_vol_trl_20d	avg=+18.4%	->	psq_fwd_5d	avg=-3.4%
+0.56	spy_vol_trl_20d	avg=+18.4%	->	bottom10_fwd_20d	avg=+7.9%
+0.55	qqq_vol_trl_20d	avg=+23.2%	->	bottom10_fwd_20d	avg=+7.9%
+0.55	spy_vol_trl_20d	avg=+18.4%	->	bottom10_fwd_5d	avg=+5.1%
-0.54	qqq_vol_trl_5d	avg=+21.8%	->	psq_fwd_5d	avg=-3.4%
+0.54	spy_vol_trl_5d	avg=+16.9%	->	bottom10_fwd_20d	avg=+7.9%
+0.54	spy_vol_trl_20d	avg=+18.4%	->	universe_fwd_5d	avg=+5.0%
+0.54	book_trl_20d	avg=+2.8%	->	spread_fwd_5d	avg=+1.0%
-0.53	spy_vol_trl_5d	avg=+16.9%	->	psq_fwd_5d	avg=-3.4%
-0.53	book_trl_20d	avg=+2.8%	->	bottom10_fwd_5d	avg=+5.1%
+0.52	universe_trl_20d	avg=+1.9%	->	spread_fwd_5d	avg=+1.0%
+0.52	spy_vol_trl_5d	avg=+16.9%	->	universe_fwd_5d	avg=+5.0%
+0.52	qqq_vol_trl_5d	avg=+21.8%	->	universe_fwd_5d	avg=+5.0%
-0.52	universe_trl_20d	avg=+1.9%	->	bottom10_fwd_5d	avg=+5.1%
+0.52	qqq_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_20d	avg=+7.9%
+0.52	spy_vol_trl_5d	avg=+16.9%	->	bottom10_fwd_5d	avg=+5.1%
+0.51	qqq_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_5d	avg=+5.1%
+0.51	qqq_trl_20d	avg=+0.8%	->	spread_fwd_5d	avg=+1.0%

UP > +6.5% (n=127)

MODERATE (0.50-0.70):

+0.66	book_trl_20d	avg=+3.9%	->	spread_fwd_5d	avg=+2.0%
-0.65	book_trl_20d	avg=+3.9%	->	bottom10_fwd_5d	avg=+5.9%
-0.65	qqq_vol_trl_5d	avg=+21.9%	->	psq_fwd_5d	avg=-3.9%
+0.65	spy_vol_trl_5d	avg=+17.7%	->	bottom10_fwd_20d	avg=+9.1%
+0.64	qqq_vol_trl_20d	avg=+25.3%	->	bottom10_fwd_20d	avg=+9.1%
+0.64	spy_vol_trl_20d	avg=+20.7%	->	bottom10_fwd_20d	avg=+9.1%
+0.63	qqq_vol_trl_5d	avg=+21.9%	->	bottom10_fwd_20d	avg=+9.1%
-0.62	qqq_vol_trl_20d	avg=+25.3%	->	psq_fwd_5d	avg=-3.9%
-0.62	spy_vol_trl_5d	avg=+17.7%	->	psq_fwd_5d	avg=-3.9%
-0.61	universe_trl_20d	avg=+2.8%	->	bottom10_fwd_5d	avg=+5.9%
+0.60	universe_trl_20d	avg=+2.8%	->	spread_fwd_5d	avg=+2.0%
+0.60	qqq_vol_trl_5d	avg=+21.9%	->	bottom10_fwd_5d	avg=+5.9%
-0.60	spy_vol_trl_20d	avg=+20.7%	->	psq_fwd_5d	avg=-3.9%
+0.60	qqq_vol_trl_20d	avg=+25.3%	->	bottom10_fwd_5d	avg=+5.9%
+0.59	spy_vol_trl_5d	avg=+17.7%	->	bottom10_fwd_5d	avg=+5.9%
+0.59	qqq_trl_20d	avg=+1.6%	->	spread_fwd_5d	avg=+2.0%
-0.58	bottom10_trl_20d	avg=+3.2%	->	bottom10_fwd_5d	avg=+5.9%
+0.58	bottom10_trl_20d	avg=+3.2%	->	spread_fwd_5d	avg=+2.0%
+0.57	qqq_vol_trl_5d	avg=+21.9%	->	universe_fwd_5d	avg=+5.8%
+0.57	spy_trl_20d	avg=+0.9%	->	spread_fwd_5d	avg=+2.0%
+0.57	spy_vol_trl_20d	avg=+20.7%	->	bottom10_fwd_5d	avg=+5.9%
-0.57	qqq_trl_20d	avg=+1.6%	->	bottom10_fwd_5d	avg=+5.9%
-0.56	spy_vol_trl_20d	avg=+20.7%	->	psq_fwd_20d	avg=-5.1%
-0.56	qqq_vol_trl_20d	avg=+25.3%	->	spread_fwd_5d	avg=+2.0%
-0.56	qqq_vol_trl_20d	avg=+25.3%	->	psq_fwd_20d	avg=-5.1%
-0.56	spy_trl_20d	avg=+0.9%	->	bottom10_fwd_5d	avg=+5.9%
+0.55	spy_vol_trl_5d	avg=+17.7%	->	universe_fwd_5d	avg=+5.8%
+0.55	top10_trl_20d	avg=+4.4%	->	spread_fwd_5d	avg=+2.0%
-0.55	spy_trl_20d	avg=+0.9%	->	bottom10_fwd_20d	avg=+9.1%
+0.55	spy_trl_20d	avg=+0.9%	->	spread_fwd_20d	avg=+1.8%
+0.55	book_trl_20d	avg=+3.9%	->	spread_fwd_20d	avg=+1.8%
-0.55	qqq_vol_trl_5d	avg=+21.9%	->	spread_fwd_5d	avg=+2.0%
-0.55	spy_vol_trl_5d	avg=+17.7%	->	psq_fwd_20d	avg=-5.1%
-0.55	book_trl_20d	avg=+3.9%	->	universe_fwd_5d	avg=+5.8%
+0.54	book_trl_20d	avg=+3.9%	->	psq_fwd_5d	avg=-3.9%
-0.54	spy_vol_trl_20d	avg=+20.7%	->	spread_fwd_5d	avg=+2.0%
+0.54	qqq_vol_trl_20d	avg=+25.3%	->	universe_fwd_5d	avg=+5.8%
-0.54	spy_vol_trl_5d	avg=+17.7%	->	spread_fwd_5d	avg=+2.0%
-0.53	book_trl_20d	avg=+3.9%	->	bottom10_fwd_20d	avg=+9.1%
+0.53	spy_trl_20d	avg=+0.9%	->	psq_fwd_5d	avg=-3.9%
-0.53	psq_trl_20d	avg=-1.5%	->	spread_fwd_5d	avg=+2.0%
+0.53	qqq_trl_20d	avg=+1.6%	->	spread_fwd_20d	avg=+1.8%
-0.53	qqq_vol_trl_5d	avg=+21.9%	->	psq_fwd_20d	avg=-5.1%
+0.53	universe_trl_20d	avg=+2.8%	->	psq_fwd_5d	avg=-3.9%
+0.52	smh_trl_20d	avg=+3.5%	->	spread_fwd_5d	avg=+2.0%
-0.52	universe_trl_20d	avg=+2.8%	->	universe_fwd_5d	avg=+5.8%
+0.52	smh_trl_20d	avg=+3.5%	->	spread_fwd_20d	avg=+1.8%
-0.52	spy_trl_20d	avg=+0.9%	->	universe_fwd_5d	avg=+5.8%
-0.51	top10_trl_20d	avg=+4.4%	->	bottom10_fwd_5d	avg=+5.9%
+0.51	spy_vol_trl_20d	avg=+20.7%	->	universe_fwd_5d	avg=+5.8%
+0.51	psq_trl_20d	avg=-1.5%	->	bottom10_fwd_5d	avg=+5.9%
+0.50	spy_vol_trl_5d	avg=+17.7%	->	universe_fwd_20d	avg=+8.3%

DOWN < -4.5% (n=222)

MODERATE (0.50-0.70):

-0.57	qqq_vol_trl_5d	avg=+22.1%	->	bottom10_fwd_5d	avg=-6.2%
-0.54	qqq_vol_trl_5d	avg=+22.1%	->	universe_fwd_5d	avg=-5.8%
-0.54	spy_vol_trl_5d	avg=+17.0%	->	bottom10_fwd_5d	avg=-6.2%

-0.53	qqq_vol_trl_20d	avg=+21.0%	->	bottom10_fwd_5d	avg=-6.2%
-0.53	qqq_vol_trl_20d	avg=+21.0%	->	universe_fwd_5d	avg=-5.8%
-0.52	spy_vol_trl_5d	avg=+17.0%	->	universe_fwd_5d	avg=-5.8%
-0.51	spy_vol_trl_20d	avg=+15.5%	->	universe_fwd_5d	avg=-5.8%
-0.51	spy_vol_trl_20d	avg=+15.5%	->	bottom10_fwd_5d	avg=-6.2%

DOWN < -6.5% (n=103)

MODERATE (0.50-0.70):

-0.66	qqq_vol_trl_20d	avg=+22.4%	->	universe_fwd_5d	avg=-7.8%
-0.66	qqq_vol_trl_20d	avg=+22.4%	->	bottom10_fwd_5d	avg=-8.5%
-0.65	spy_vol_trl_20d	avg=+16.8%	->	universe_fwd_5d	avg=-7.8%
-0.65	qqq_vol_trl_5d	avg=+24.5%	->	bottom10_fwd_5d	avg=-8.5%
-0.64	spy_vol_trl_20d	avg=+16.8%	->	bottom10_fwd_5d	avg=-8.5%
-0.62	qqq_vol_trl_5d	avg=+24.5%	->	universe_fwd_5d	avg=-7.8%
-0.62	spy_vol_trl_5d	avg=+19.5%	->	bottom10_fwd_5d	avg=-8.5%
-0.60	qqq_vol_trl_20d	avg=+22.4%	->	top10_fwd_5d	avg=-8.1%
-0.60	spy_vol_trl_5d	avg=+19.5%	->	universe_fwd_5d	avg=-7.8%
-0.56	spy_vol_trl_20d	avg=+16.8%	->	top10_fwd_5d	avg=-8.1%
-0.56	spy_vol_trl_1d	avg=+18.2%	->	bottom10_fwd_5d	avg=-8.5%
+0.55	book_trl_5d	avg=+0.2%	->	spread_fwd_20d	avg=-3.6%
-0.55	spy_vol_trl_20d	avg=+16.8%	->	smh_fwd_5d	avg=-8.1%
-0.54	spy_vol_trl_1d	avg=+18.2%	->	universe_fwd_5d	avg=-7.8%
-0.53	qqq_vol_trl_5d	avg=+24.5%	->	bottom10_fwd_1d	avg=-1.9%
+0.53	spy_trl_5d	avg=-0.3%	->	spread_fwd_20d	avg=-3.6%
-0.53	spy_vol_trl_5d	avg=+19.5%	->	bottom10_fwd_1d	avg=-1.9%
-0.53	qqq_vol_trl_20d	avg=+22.4%	->	smh_fwd_5d	avg=-8.1%
-0.52	qqq_vol_trl_1d	avg=+22.2%	->	bottom10_fwd_5d	avg=-8.5%
-0.52	spy_vol_trl_5d	avg=+19.5%	->	smh_fwd_5d	avg=-8.1%
-0.51	qqq_vol_trl_5d	avg=+24.5%	->	smh_fwd_5d	avg=-8.1%

Forward 20-Day portfolio return (filtered by book_fwd_20d)

UP > +7.5% (n=655)

MODERATE (0.50-0.70):

+0.68	qqq_vol_trl_20d	avg=+21.6%	->	bottom10_fwd_20d	avg=+9.9%
+0.67	spy_vol_trl_20d	avg=+17.5%	->	bottom10_fwd_20d	avg=+9.9%
+0.65	spy_vol_trl_5d	avg=+16.5%	->	bottom10_fwd_20d	avg=+9.9%
+0.64	qqq_vol_trl_5d	avg=+20.3%	->	bottom10_fwd_20d	avg=+9.9%
-0.61	spy_vol_trl_20d	avg=+17.5%	->	psq_fwd_20d	avg=-5.7%
-0.60	spy_vol_trl_5d	avg=+16.5%	->	psq_fwd_20d	avg=-5.7%
-0.59	qqq_vol_trl_20d	avg=+21.6%	->	psq_fwd_20d	avg=-5.7%
+0.59	qqq_vol_trl_20d	avg=+21.6%	->	universe_fwd_20d	avg=+9.4%
+0.58	spy_vol_trl_20d	avg=+17.5%	->	universe_fwd_20d	avg=+9.4%
+0.57	spy_vol_trl_5d	avg=+16.5%	->	universe_fwd_20d	avg=+9.4%
+0.57	qqq_vol_trl_5d	avg=+20.3%	->	universe_fwd_20d	avg=+9.4%
-0.56	qqq_vol_trl_5d	avg=+20.3%	->	psq_fwd_20d	avg=-5.7%

UP > +12.5% (n=268)

STRONG (|r|>=0.70):

+0.76	spy_vol_trl_5d	avg=+19.7%	->	bottom10_fwd_20d	avg=+12.2%
+0.75	qqq_vol_trl_5d	avg=+24.0%	->	bottom10_fwd_20d	avg=+12.2%
+0.75	qqq_vol_trl_20d	avg=+24.6%	->	bottom10_fwd_20d	avg=+12.2%
+0.74	spy_vol_trl_20d	avg=+20.2%	->	bottom10_fwd_20d	avg=+12.2%
-0.73	spy_trl_20d	avg=+0.2%	->	bottom10_fwd_20d	avg=+12.2%

MODERATE (0.50-0.70):

-0.66	spy_vol_trl_20d	avg=+20.2%	->	psq_fwd_20d	avg=-7.0%
+0.66	spy_trl_20d	avg=+0.2%	->	psq_fwd_20d	avg=-7.0%
-0.66	universe_trl_20d	avg=+1.2%	->	bottom10_fwd_20d	avg=+12.2%
-0.65	qqq_vol_trl_20d	avg=+24.6%	->	psq_fwd_20d	avg=-7.0%
-0.65	spy_vol_trl_5d	avg=+19.7%	->	psq_fwd_20d	avg=-7.0%
+0.65	qqq_vol_trl_5d	avg=+24.0%	->	universe_fwd_20d	avg=+11.9%

-0.65	bottom10_trl_20d	avg=+1.7%	->	bottom10_fwd_20d	avg=+12.2%
+0.65	spy_vol_trl_5d	avg=+19.7%	->	universe_fwd_20d	avg=+11.9%
+0.63	universe_trl_20d	avg=+1.2%	->	psq_fwd_20d	avg=-7.0%
-0.62	book_trl_20d	avg=+1.8%	->	bottom10_fwd_20d	avg=+12.2%
+0.62	spy_trl_20d	avg=+0.2%	->	spread_fwd_20d	avg=+3.6%
+0.62	qqq_vol_trl_20d	avg=+24.6%	->	universe_fwd_20d	avg=+11.9%
-0.62	qqq_vol_trl_5d	avg=+24.0%	->	psq_fwd_20d	avg=-7.0%
+0.62	spy_vol_trl_20d	avg=+20.2%	->	universe_fwd_20d	avg=+11.9%
-0.61	qqq_trl_20d	avg=+0.6%	->	bottom10_fwd_20d	avg=+12.2%
-0.60	spy_vol_trl_5d	avg=+19.7%	->	spread_fwd_20d	avg=+3.6%
+0.59	book_trl_20d	avg=+1.8%	->	spread_fwd_20d	avg=+3.6%
-0.59	spy_trl_20d	avg=+0.2%	->	universe_fwd_20d	avg=+11.9%
+0.59	qqq_trl_20d	avg=+0.6%	->	psq_fwd_20d	avg=-7.0%
-0.59	qqq_vol_trl_20d	avg=+24.6%	->	spread_fwd_20d	avg=+3.6%
-0.59	qqq_vol_trl_5d	avg=+24.0%	->	spread_fwd_20d	avg=+3.6%
+0.58	psq_trl_20d	avg=-0.4%	->	bottom10_fwd_20d	avg=+12.2%
-0.58	spy_vol_trl_20d	avg=+20.2%	->	spread_fwd_20d	avg=+3.6%
-0.57	universe_trl_20d	avg=+1.2%	->	universe_fwd_20d	avg=+11.9%
+0.57	top10_trl_20d	avg=+1.5%	->	spread_fwd_20d	avg=+3.6%
-0.56	psq_trl_20d	avg=-0.4%	->	psq_fwd_20d	avg=-7.0%
+0.55	bottom10_trl_20d	avg=+1.7%	->	psq_fwd_20d	avg=-7.0%
+0.55	qqq_trl_20d	avg=+0.6%	->	spread_fwd_20d	avg=+3.6%
+0.54	spy_vol_trl_1d	avg=+15.2%	->	bottom10_fwd_20d	avg=+12.2%
-0.53	qqq_trl_20d	avg=+0.6%	->	universe_fwd_20d	avg=+11.9%
+0.53	book_trl_20d	avg=+1.8%	->	psq_fwd_20d	avg=-7.0%
-0.52	bottom10_trl_20d	avg=+1.7%	->	universe_fwd_20d	avg=+11.9%
+0.52	spy_trl_20d	avg=+0.2%	->	spread_fwd_5d	avg=+0.4%
-0.52	spy_trl_20d	avg=+0.2%	->	bottom10_fwd_5d	avg=+3.6%
-0.51	psq_trl_20d	avg=-0.4%	->	spread_fwd_20d	avg=+3.6%
+0.51	smh_trl_20d	avg=+1.4%	->	spread_fwd_20d	avg=+3.6%
+0.51	universe_trl_20d	avg=+1.2%	->	spread_fwd_20d	avg=+3.6%
+0.50	psq_trl_20d	avg=-0.4%	->	universe_fwd_20d	avg=+11.9%
-0.50	book_trl_20d	avg=+1.8%	->	universe_fwd_20d	avg=+11.9%

DOWN < -7.5% (n=215)

(none |r|>=0.5)

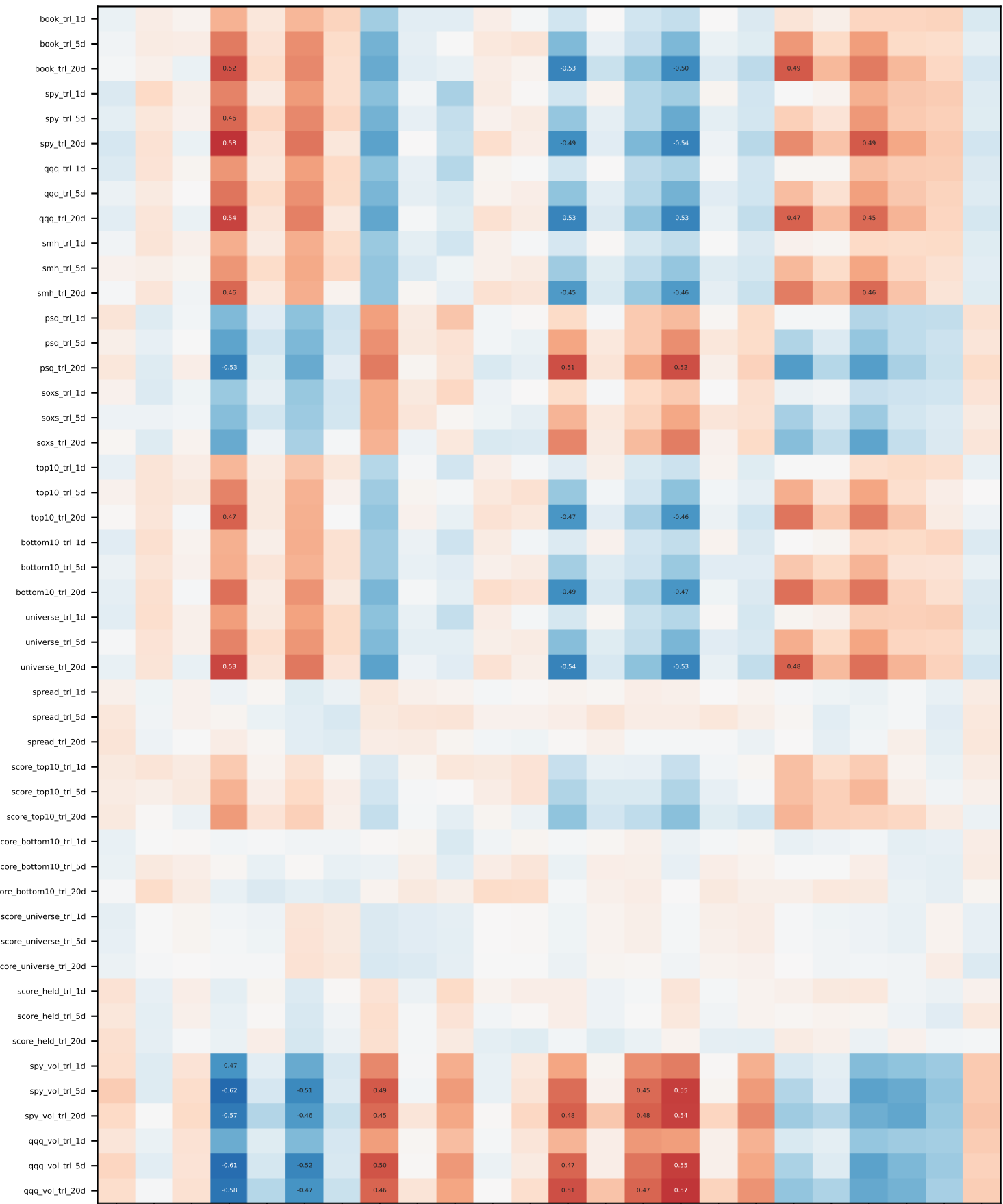
DOWN < -12.5% (n=70)

MODERATE (0.50-0.70):

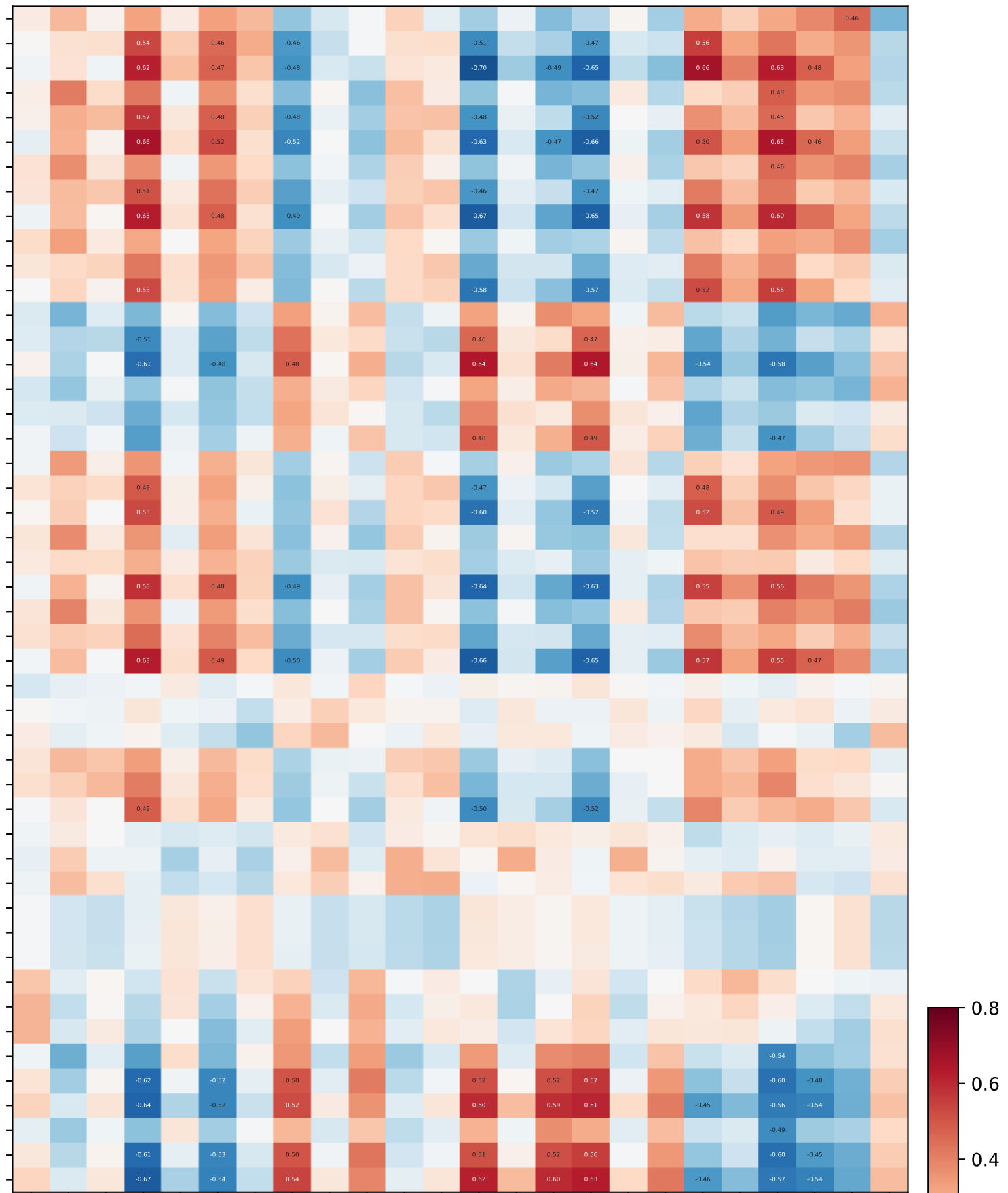
-0.58	psq_trl_20d	avg=+0.1%	->	spread_fwd_20d	avg=-1.2%
-0.57	spy_vol_trl_5d	avg=+15.7%	->	bottom10_fwd_5d	avg=-3.6%
+0.57	book_trl_20d	avg=-0.1%	->	spread_fwd_20d	avg=-1.2%
-0.57	spy_vol_trl_5d	avg=+15.7%	->	soxs_fwd_20d	avg=+35.9%
-0.56	qqq_vol_trl_5d	avg=+20.3%	->	soxs_fwd_20d	avg=+35.9%
+0.55	spy_trl_20d	avg=-0.2%	->	soxs_fwd_20d	avg=+35.9%
+0.55	qqq_trl_20d	avg=+0.3%	->	spread_fwd_20d	avg=-1.2%
-0.54	spy_vol_trl_20d	avg=+14.8%	->	bottom10_fwd_5d	avg=-3.6%
-0.53	qqq_vol_trl_20d	avg=+20.2%	->	soxs_fwd_20d	avg=+35.9%
-0.52	spy_vol_trl_20d	avg=+14.8%	->	soxs_fwd_20d	avg=+35.9%
-0.52	qqq_vol_trl_5d	avg=+20.3%	->	bottom10_fwd_5d	avg=-3.6%
+0.52	book_trl_5d	avg=+0.2%	->	spread_fwd_20d	avg=-1.2%
-0.51	score_top10_trl_1d	avg=0.831	->	book_fwd_20d	avg=-17.7%
-0.51	soxs_trl_20d	avg=+1.5%	->	spread_fwd_20d	avg=-1.2%
+0.50	universe_trl_20d	avg=+0.6%	->	spread_fwd_20d	avg=-1.2%

Trailing (rows) vs Forward (cols) correlations — filtered by Forward 1-Day portfolio return | scale +/-0.8

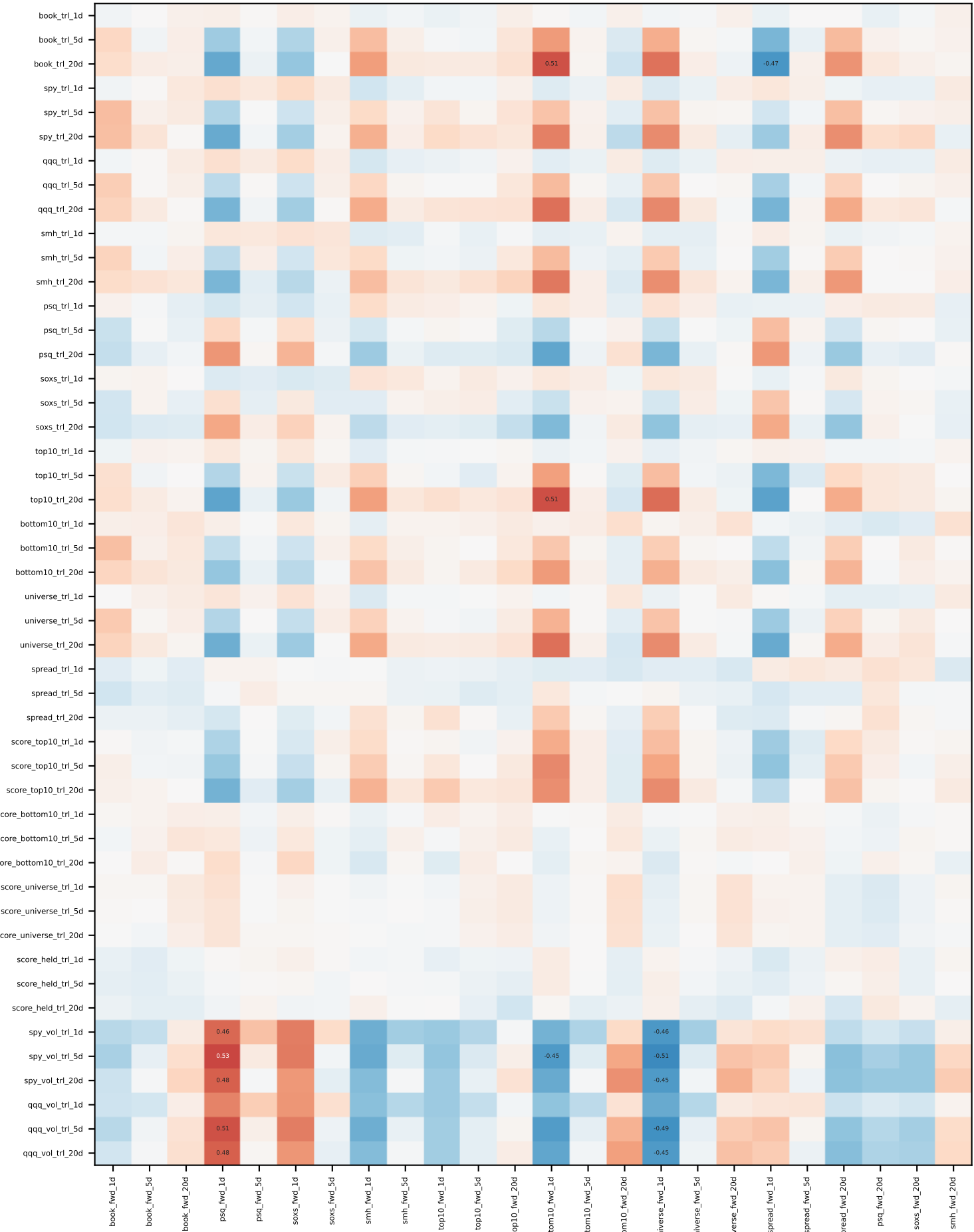
UP > +2.0% (n=296)



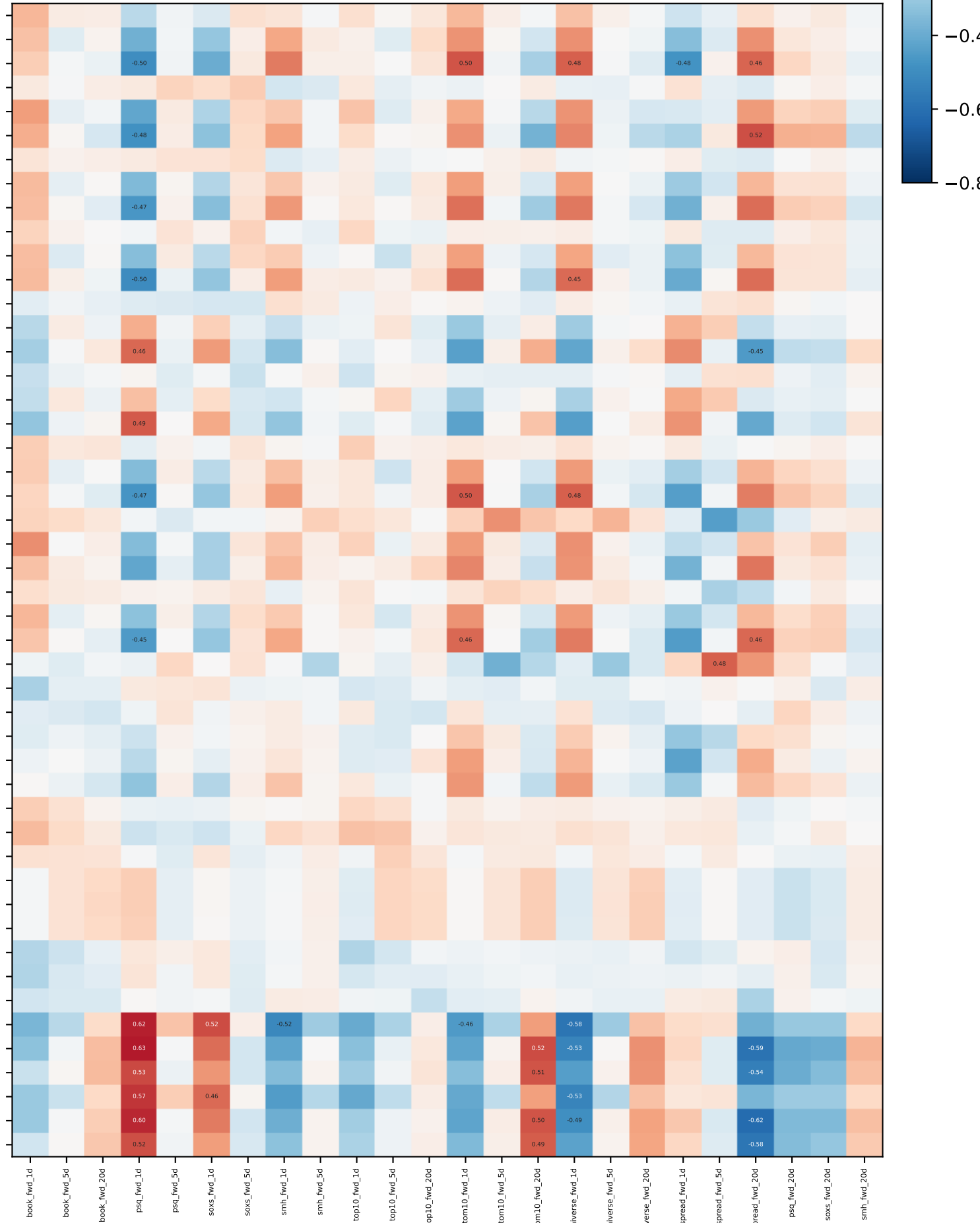
UP > +3.5% (n=68)



DOWN < -2.0% (n=250)

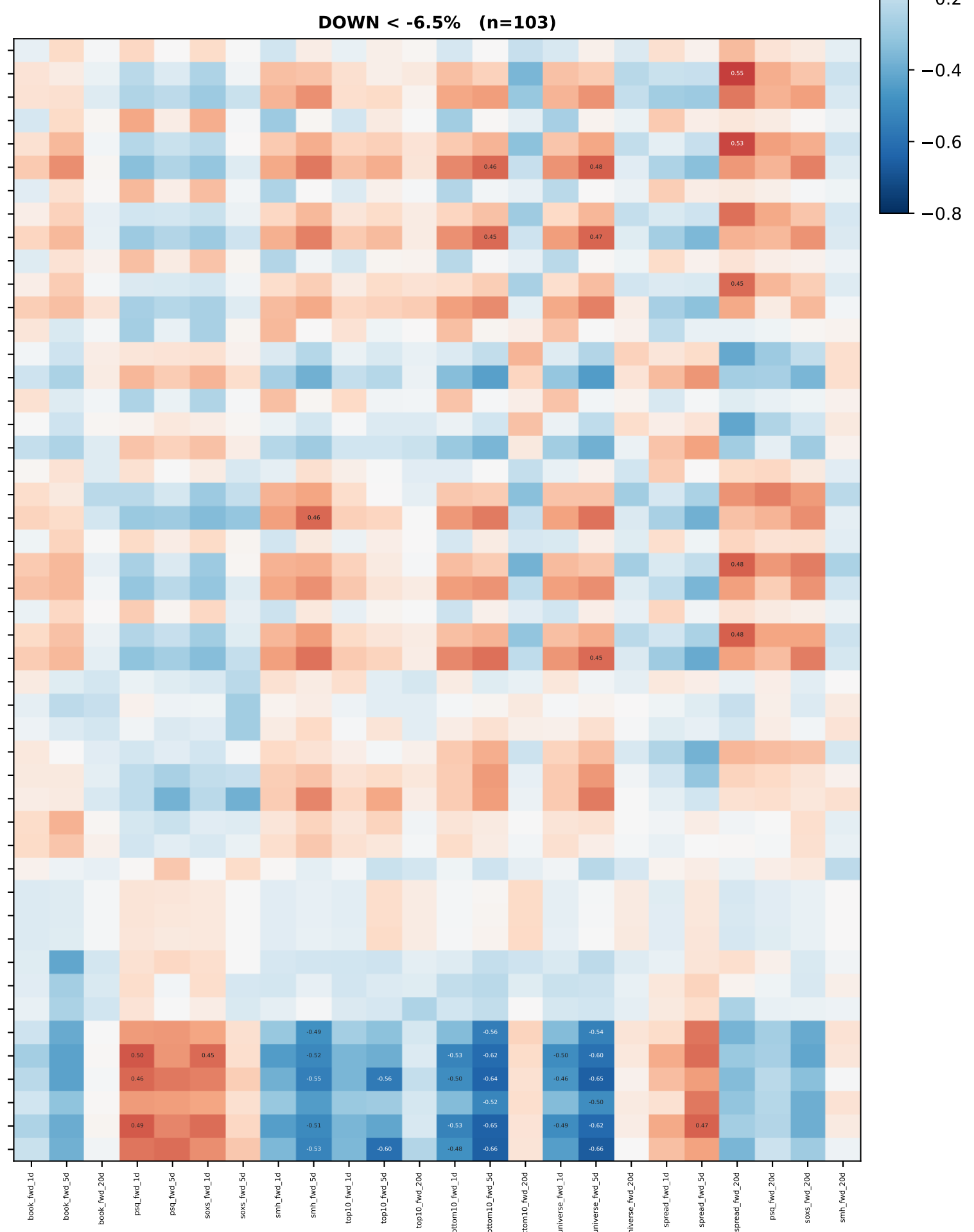
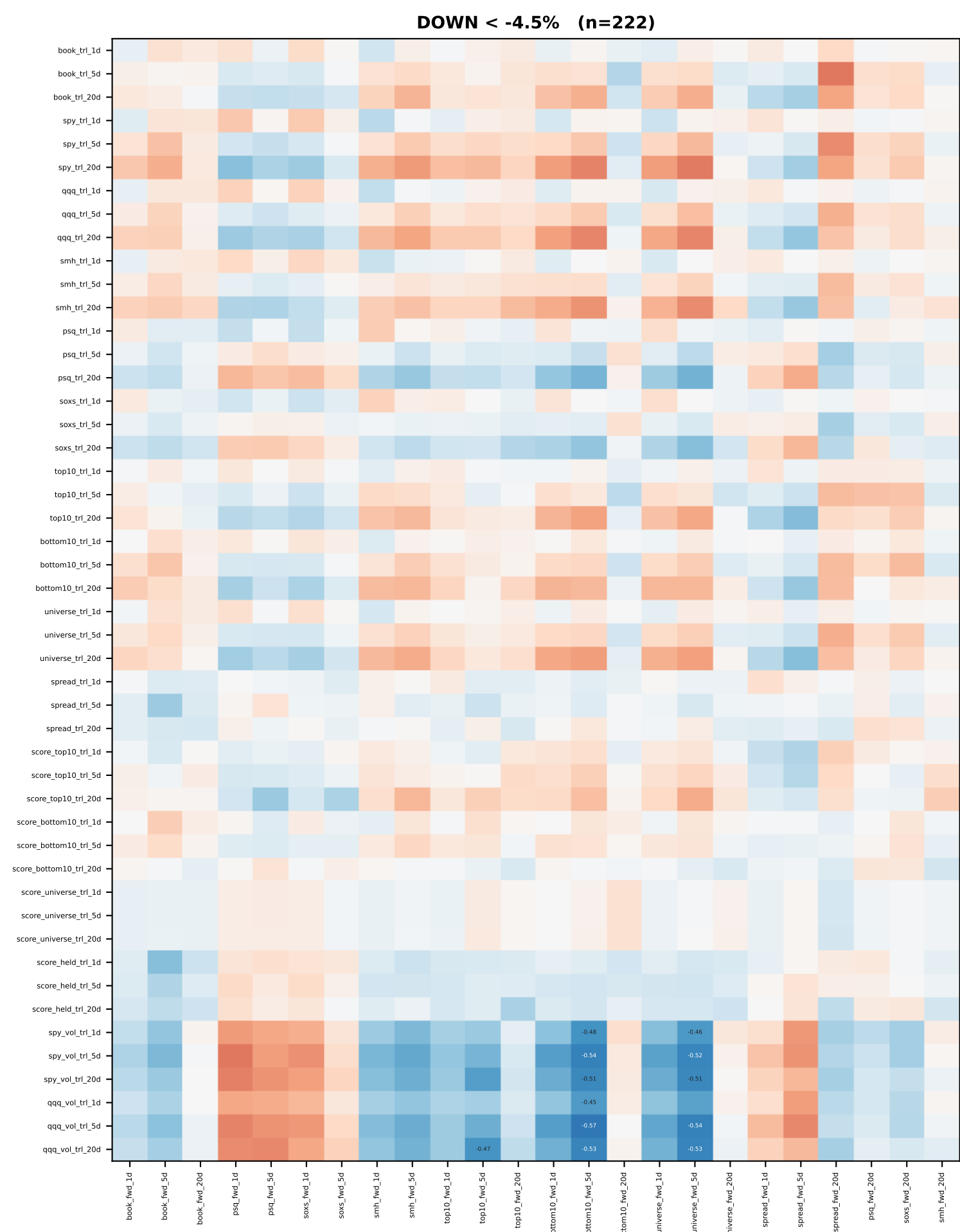
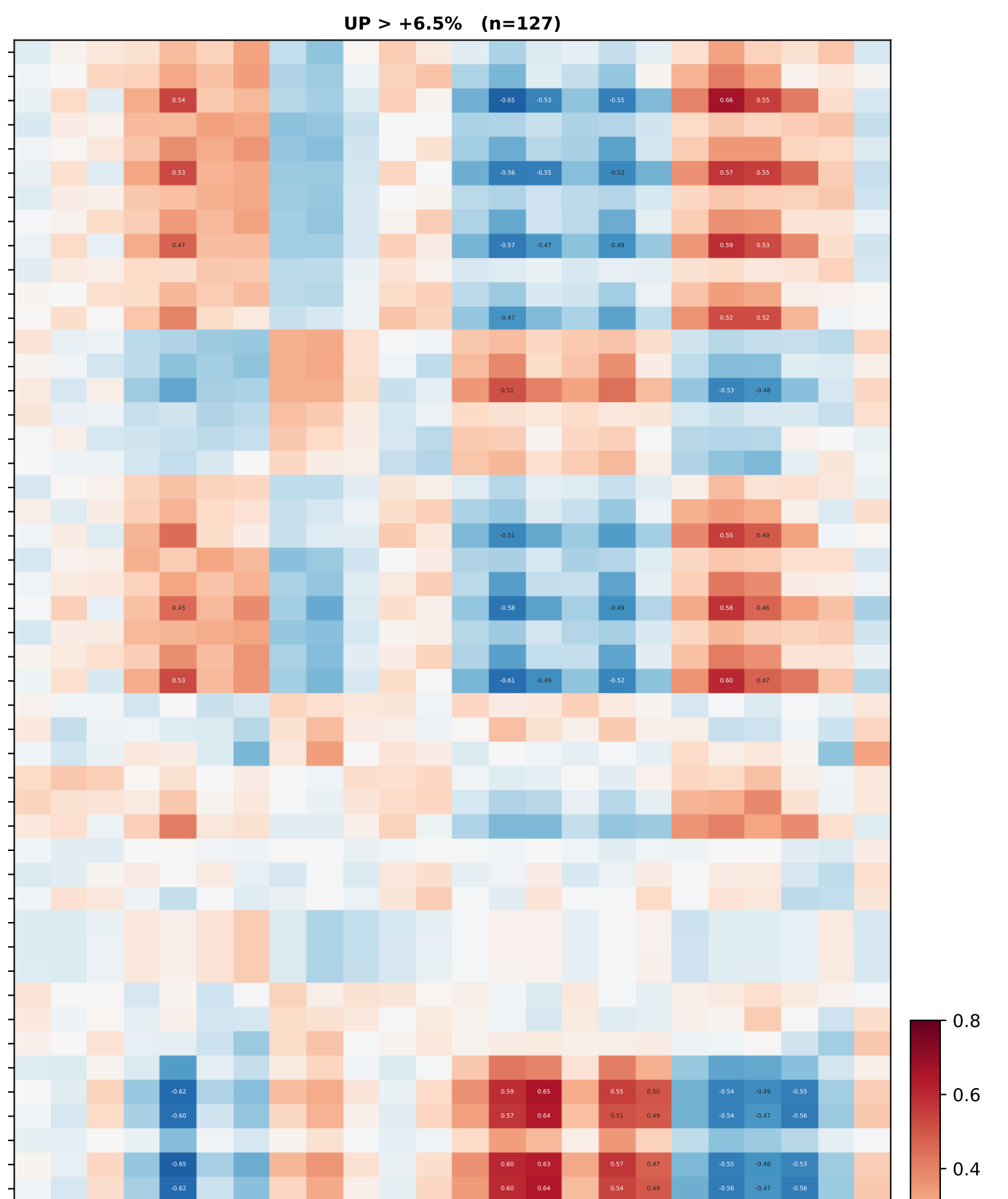
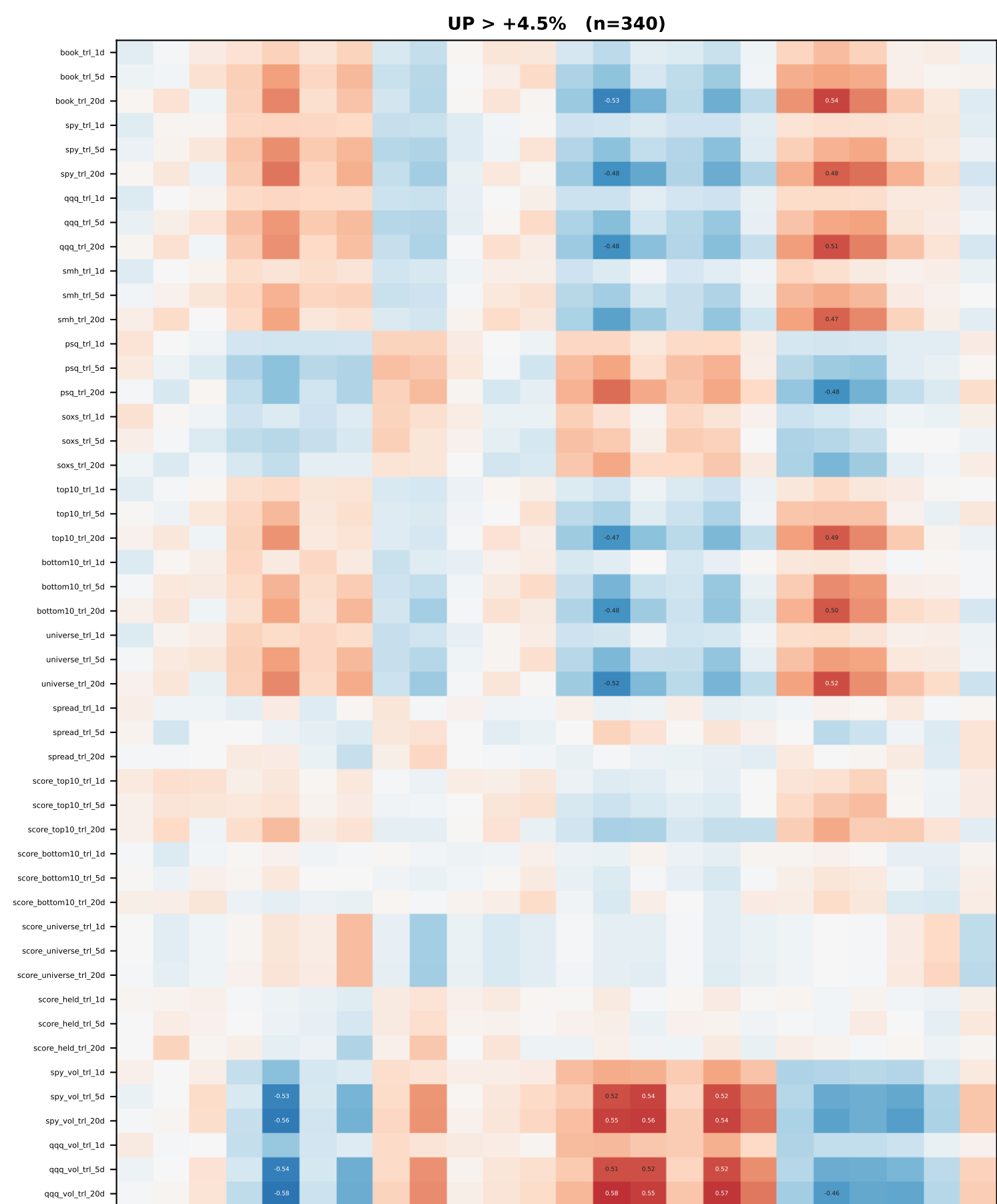


DOWN < -3.5% (n=77)

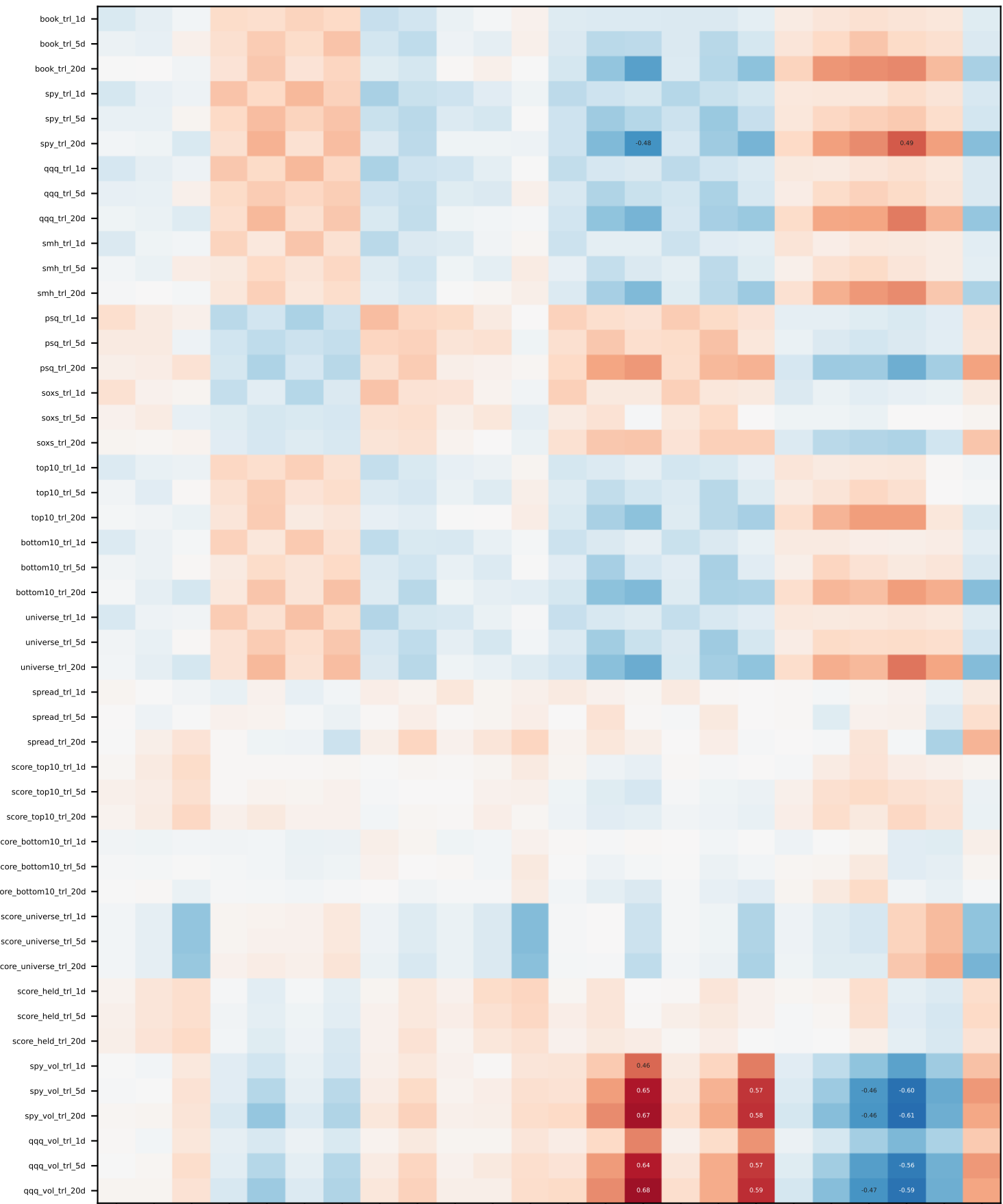


Pearson r

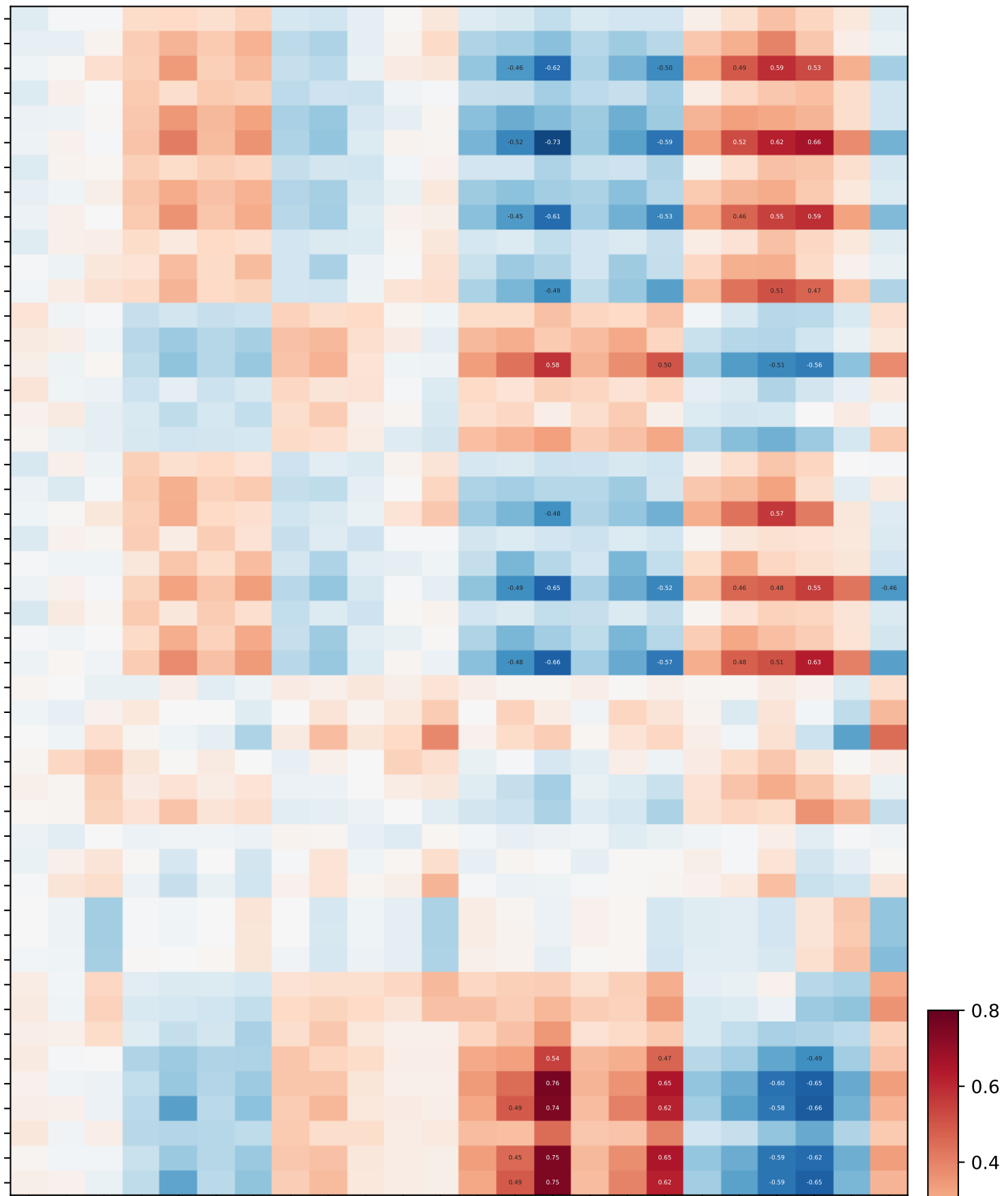
Trailing (rows) vs Forward (cols) correlations — filtered by Forward 5-Day portfolio return | scale +/-0.8



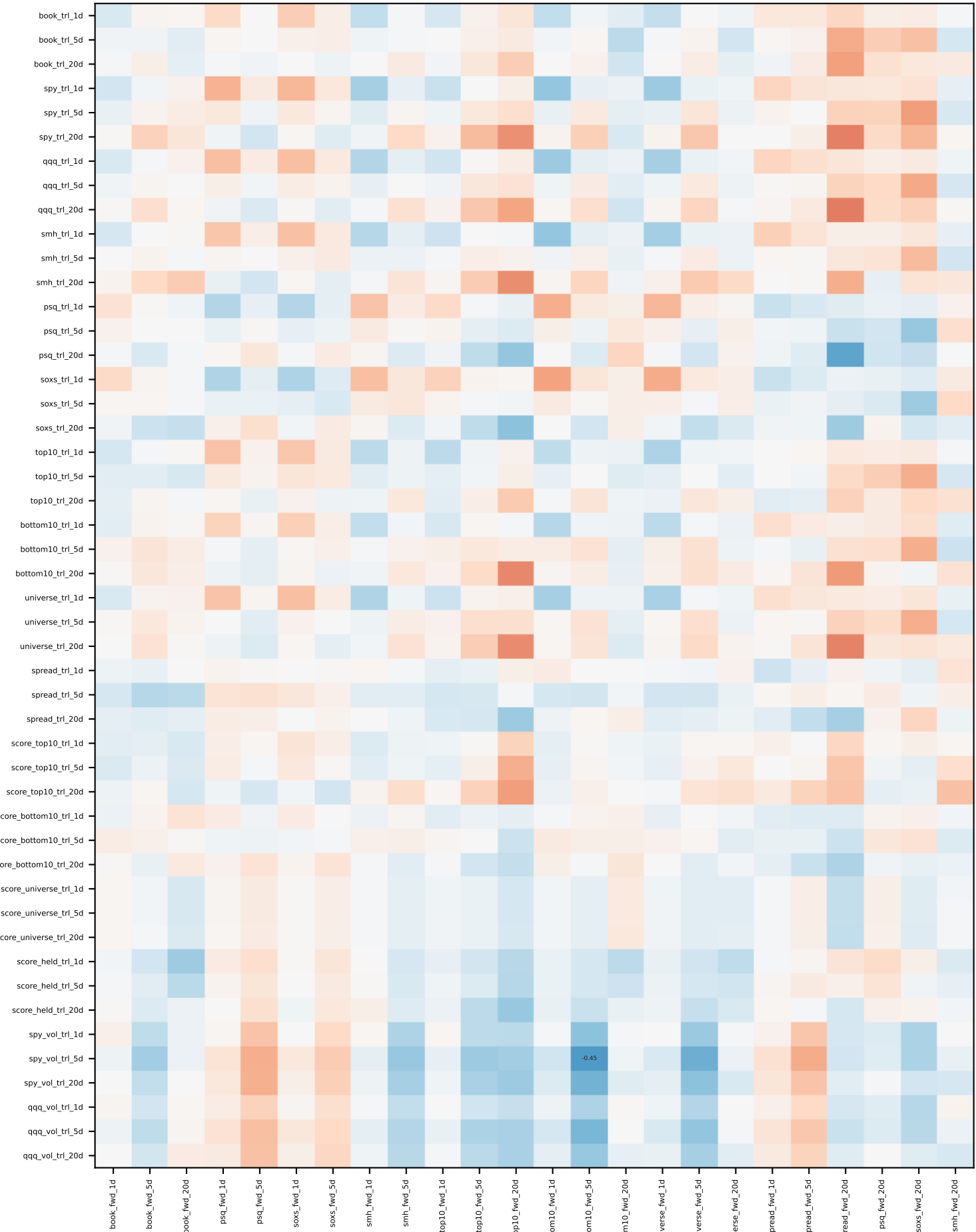
UP > +7.5% (n=655)



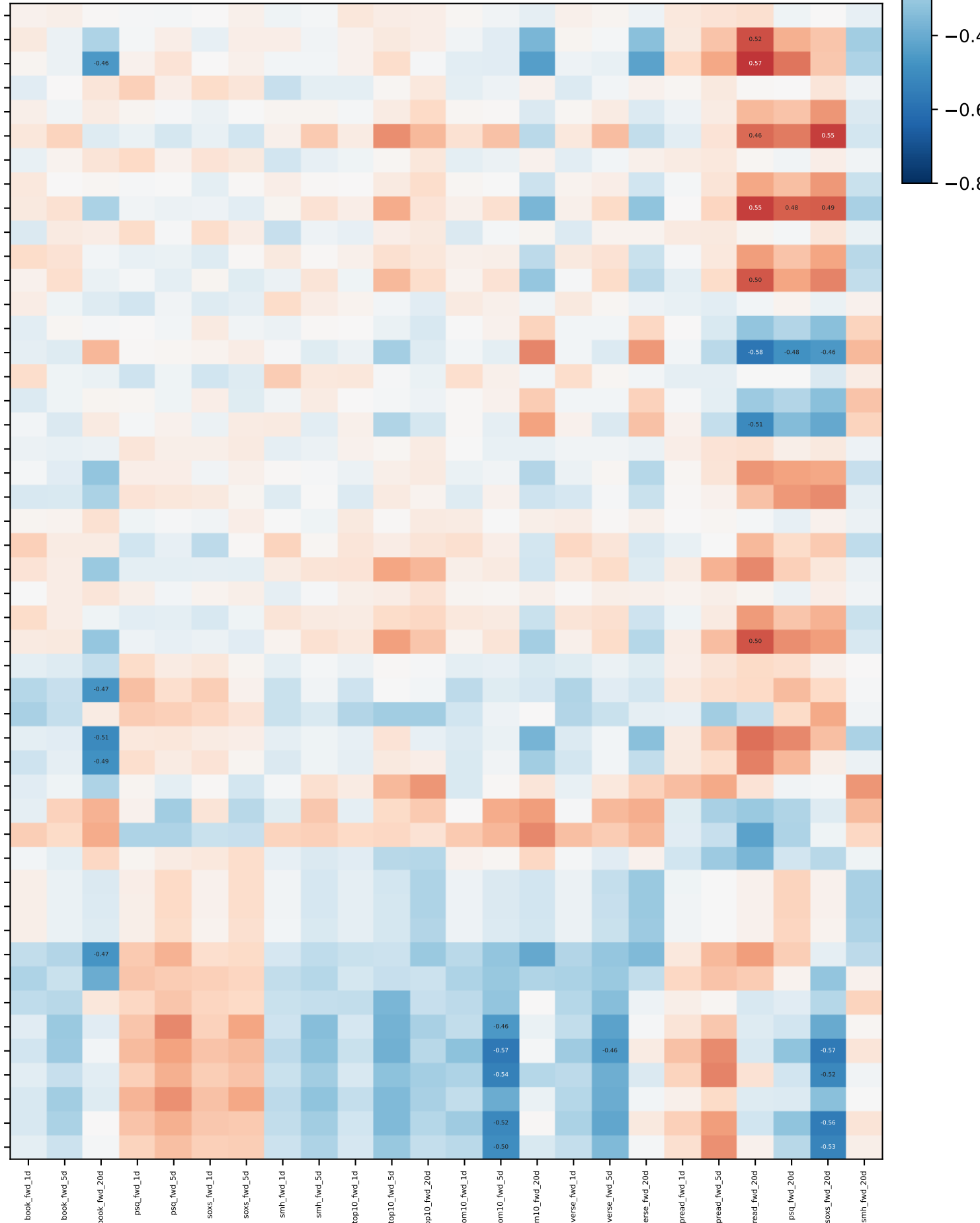
UP > +12.5% (n=268)



DOWN < -7.5% (n=215)



DOWN < -12.5% (n=70)



Pearson r

Appendix B

Correlation & Gap Analysis (since-COVID, vol-normalized)

(source: corr_tails_by_horizon_since_covid_volz.pdf)

D = UP minus DN. Returns/vol abs in %; scores raw.

JUN-15 z shows how far that factor sat above/below its recent norm at that close; compare its sign/size to the UP vs DN segment z to read the regime.

Biggest UP-DN gaps, sorted by |z-gap| within each horizon

SINCE COVID (2020-04-01 to 2026-06-18, n=1562) | z = rolling z (value-recent mean)/recent std, L=63d | JUN-15 = 2026-06-15 close

Sorted by absolute normalized gap. Dz colored; (abs gap | zUP/zDN) grey. JUN15 = factor z at 2026-06-15.

FWD 1D (thr +/-2.0%)

factor	Dz15z	(absGap zU/zD)
1.sc.held_20d	+0.20.00	+0.004 +0.28/+0.08
2.spy_vol_5d	-0.11.46	-0.6 +0.03/+0.21
3.sc.universe_20d	+0.10.51	+0.000 -0.57/-0.74
4.spy_1d	-0.11.65	-0.2 -0.15/+0.01
5.psq_1d	+0.11.98	+0.2 +0.17/+0.01
6.qqq_vol_5d	-0.12.20	-0.9 +0.08/+0.23
7.qqq_1d	-0.12.02	-0.2 -0.16/-0.02
8.soxs_1d	+0.11.40	+1.0 +0.16/+0.02
9.smh_1d	-0.11.30	-0.3 -0.16/-0.02
10.soxs_5d	+0.11.07	+2.1 +0.17/+0.04
11.sc.bottom10_20d	+0.10.38	+0.001 +0.09/-0.02
12.sc.universe_5d	+0.10.80	+0.000 -0.40/-0.50
13.sc.top10_20d	+0.10.23	+0.001 +0.07/-0.03
14.smh_5d	-0.10.84	-0.5 -0.17/-0.08
15.book_5d	-0.10.49	-0.4 -0.22/-0.12
16.spread_1d	-0.01.20	-0.2 -0.14/-0.05
17.book_20d	+0.00.07	+0.9 +0.01/-0.08
18.psq_5d	+0.00.82	+0.2 +0.22/+0.14
19.universe_20d	+0.00.28	+0.4 -0.14/-0.23
20.spy_20d	+0.00.18	+0.1 -0.25/-0.33
21.top10_5d	-0.00.46	-0.3 -0.16/+0.07
22.smh_20d	+0.00.21	+0.3 -0.10/+0.18
23.qqq_5d	-0.00.78	-0.2 -0.22/-0.14
24.sc.bottom10_5d	+0.01.28	+0.002 -0.02/-0.09
25.universe_1d	-0.01.43	-0.1 -0.16/-0.09
26.sc.held_1d	+0.00.30	+0.002 +0.12/+0.05
27.bottom10_20d	+0.00.22	+0.5 -0.13/-0.19
28.spread_5d	-0.00.70	-0.2 +0.00/+0.06
29.soxs_20d	-0.00.71	-1.1 +0.10/+0.15
30.sc.held_5d	+0.01.09	+0.001 +0.14/+0.09
31.qqq_vol_1d	+0.02.22	+0.5 +0.09/+0.05
32.spread_20d	-0.00.50	+0.2 +0.01/+0.05
33.spy_5d	-0.00.61	-0.1 -0.22/-0.18
34.top10_1d	-0.01.72	-0.1 -0.15/-0.12
35.sc.universe_1d	-0.01.26	+0.000 -0.41/-0.38
36.qqq_20d	+0.00.15	+0.1 -0.23/-0.27
37.psq_20d	-0.00.04	-0.2 +0.24/+0.28
38.qqq_vol_20d	-0.02.52	+1.2 +0.17/+0.20
39.universe_5d	-0.00.10	-0.2 -0.23/-0.20
40.sc.bottom10_1d	-0.01.35	+0.001 -0.03/-0.00
41.book_1d	-0.01.26	-0.2 -0.11/-0.09
42.spy_vol_20d	-0.00.51	+1.5 +0.07/+0.08
43.bottom10_5d	-0.00.29	-0.1 -0.22/-0.21
44.sc.top10_1d	-0.00.41	-0.000 +0.03/+0.04
45.sc.top10_5d	-0.01.03	+0.001 +0.01/+0.02
46.top10_20d	+0.00.24	+0.7 -0.04/-0.05
47.spy_vol_1d	-0.01.62	-0.1 +0.07/+0.07
48.bottom10_1d	-0.00.56	+0.1 -0.06/-0.06

FWD 5D (thr +/-4.5%)

factor	Dz15z	(absGap zU/zD)
1.sc.top10_20d	+0.40.23	+0.002 +0.10/-0.30
2.book_20d	+0.34.07	+1.8 +0.13/-0.21
3.sc.held_20d	+0.33.00	+0.006 +0.32/-0.01
4.universe_20d	+0.26.28	+0.5 -0.01/-0.27
5.sc.universe_20d	+0.20.51	+0.000 -0.58/-0.82
6.sc.universe_5d	+0.20.80	+0.000 -0.39/-0.62
7.sc.held_1d	+0.20.30	+0.012 +0.18/-0.04
8.spy_vol_5d	-0.21.46	+1.5 +0.07/+0.29
9.sc.held_5d	+0.21.09	+0.009 +0.24/+0.03
10.top10_20d	+0.20.24	+1.5 +0.10/-0.10
11.sc.top10_5d	+0.13.03	+0.003 +0.03/-0.15
12.smh_20d	+0.15.21	+0.7 -0.09/-0.24
13.sc.bottom10_20d	+0.10.38	+0.001 +0.15/-0.00
14.bottom10_20d	+0.10.22	+0.3 -0.10/-0.24
15.spread_20d	+0.10.50	+1.2 +0.15/+0.01
16.sc.top10_1d	+0.10.41	+0.003 +0.02/-0.12
17.spy_20d	+0.10.18	-0.0 -0.15/-0.29
18.book_1d	-0.11.26	-0.2 -0.10/+0.03
19.qqq_vol_5d	-0.12.20	+1.1 +0.13/+0.26
20.qqq_20d	+0.10.15	+0.1 -0.13/-0.26
21.spy_vol_1d	-0.12.62	+0.7 +0.06/+0.18
22.sc.bottom10_1d	-0.12.35	-0.002 +0.10/+0.22
23.psq_20d	-0.10.04	-0.1 +0.15/+0.26
24.spy_1d	-0.11.65	-0.1 -0.07/+0.04
25.sc.universe_1d	+0.11.26	+0.000 -0.28/-0.39
26.psq_1d	+0.01.98	+0.1 +0.05/-0.04
27.qqq_1d	-0.02.02	-0.1 -0.05/+0.03
28.spread_5d	+0.00.70	+0.3 +0.06/-0.01
29.spy_5d	-0.00.61	-0.3 -0.17/-0.10
30.top10_1d	-0.01.72	-0.1 -0.04/+0.03
31.book_5d	-0.00.49	-0.2 -0.19/-0.12
32.soxs_5d	+0.01.07	+1.2 +0.12/+0.07
33.soxs_1d	+0.01.40	+0.3 -0.02/-0.07
34.spread_1d	-0.01.20	-0.1 +0.03/+0.08
35.spy_vol_20d	+0.00.51	+3.9 +0.13/+0.08
36.smh_1d	-0.01.30	-0.1 +0.01/+0.06
37.universe_1d	-0.01.43	-0.1 -0.05/-0.00
38.psq_5d	+0.00.82	+0.3 +0.16/+0.12
39.sc.bottom10_5d	-0.01.28	+0.000 +0.11/+0.15
40.qqq_vol_1d	-0.02.22	+0.7 +0.09/+0.13
41.qqq_5d	-0.00.78	-0.3 -0.15/-0.12
42.bottom10_5d	+0.00.29	-0.5 -0.14/-0.17
43.top10_5d	-0.00.46	-0.3 -0.09/-0.06
44.universe_5d	+0.00.10	-0.4 -0.14/-0.16
45.soxs_20d	-0.00.71	-1.0 +0.16/+0.18
46.qqq_vol_20d	+0.02.52	+2.8 +0.20/+0.19
47.bottom10_1d	+0.00.56	+0.0 -0.08/-0.09
48.smh_5d	-0.00.84	-0.2 -0.10/-0.10

FWD 20D (thr +/-7.5%)

factor	Dz15z	(absGap zU/zD)
1.book_20d	+0.54.07	+3.5 +0.25/-0.29
2.sc.top10_5d	+0.46.03	+0.008 +0.11/-0.35
3.sc.top10_20d	+0.42.23	+0.001 +0.10/-0.33
4.sc.held_5d	+0.31.09	+0.005 +0.14/-0.17
5.sc.top10_1d	+0.30.41	+0.007 +0.11/-0.18
6.sc.held_1d	+0.20.30	+0.007 +0.14/-0.11
7.universe_20d	+0.20.28	+1.2 +0.12/-0.13
8.spy_vol_20d	+0.20.51	+2.8 +0.11/-0.12
9.sc.bottom10_20d	+0.20.38	+0.002 +0.22/+0.00
10.top10_20d	+0.21.24	+2.6 +0.26/+0.05
11.sc.held_20d	+0.20.00	-0.007 +0.18/-0.02
12.bottom10_20d	+0.10.22	+2.3 +0.03/-0.15
13.sc.universe_20d	-0.10.51	+0.000 -0.61/-0.44
14.smh_20d	+0.10.21	+1.4 -0.04/-0.20
15.spy_vol_5d	+0.11.46	+2.8 +0.00/-0.14
16.sc.universe_5d	-0.10.80	+0.000 -0.46/-0.33
17.spread_5d	+0.10.70	+0.7 +0.14/+0.01
18.book_5d	+0.10.49	+0.5 +0.13/+0.00
19.psq_20d	-0.10.04	-0.8 +0.04/+0.14
20.sc.bottom10_5d	+0.11.28	+0.002 +0.13/+0.03
21.qqq_20d	+0.00.15	+0.7 -0.03/-0.12
22.soxs_5d	+0.01.07	+0.5 +0.02/-0.07
23.bottom10_5d	-0.00.29	-0.4 -0.04/+0.03
24.sc.bottom10_1d	-0.01.35	-0.001 +0.07/+0.14
25.spy_5d	-0.00.61	-0.1 -0.01/+0.06
26.bottom10_1d	+0.00.56	+0.1 +0.01/-0.06
27.spy_vol_1d	+0.01.62	+2.2 +0.03/-0.03
28.spread_1d	-0.01.20	-0.1 +0.07/+0.12
29.universe_1d	+0.01.43	+0.0 +0.05/-0.00
30.qqq_5d	-0.00.78	-0.1 -0.02/+0.02
31.psq_5d	+0.00.82	+0.1 +0.02/-0.02
32.sc.universe_1d	-0.01.26	+0.000 -0.36/-0.32
33.smh_5d	-0.00.84	+0.0 +0.01/+0.05
34.qqq_vol_20d	+0.02.52	+0.8 +0.16/+0.12
35.smh_1d	+0.01.30	+0.1 +0.05/+0.02
36.soxs_20d	-0.00.71	-2.9 +0.10/+0.13
37.qqq_vol_1d	-0.02.22	+0.1 +0.01/+0.03
38.book_1d	+0.01.26	+0.1 +0.09/+0.06
39.universe_5d	-0.00.10	-0.3 +0.03/+0.05
40.spy_20d	+0.00.18	+0.5 -0.02/-0.04
41.top10_5d	+0.00.46	+0.2 +0.14/+0.12
42.soxs_1d	-0.01.40	-0.1 -0.04/-0.02
43.spy_1d	+0.01.65	+0.0 +0.02/+0.00
44.psq_1d	-0.01.98	-0.0 -0.02/+0.00
45.qqq_1d	+0.02.02	+0.0 +0.02/+0.00
46.top10_1d	-0.01.72	-0.0 +0.09/+0.11
47.spread_20d	-0.00.50	+0.4 +0.07/+0.08
48.qqq_vol_5d	-0.02.20	+0.6 -0.01/-0.01

Normalized trailing(z) -> raw forward correlation call-outs (|r| >= 0.5)

SINCE COVID (2020-04-01 to 2026-06-18, n=1562) | z = rolling z (value-recent mean)/recent std, L=63d | JUN-15 = 2026-06-15 close

z=mean trailing z in panel; J15=factor z at 2026-06-15; avg=mean raw forward return. Small-n (n<~40) noisy.

Forward 1-Day portfolio return (filtered by book_fwd_1d)

UP > +2.0% (n=230)

(none |r|>=0.5)

UP > +3.5% (n=52)

MODERATE (0.50-0.70):

+0.55	universe_5d	z=-0.26	J15=-0.10	-> spread_fwd_1d	avg=+0.6%
-0.53	soxs_5d	z=+0.12	J15=-1.07	-> spread_fwd_1d	avg=+0.6%
+0.52	book_5d	z=-0.10	J15=+0.49	-> spread_fwd_1d	avg=+0.6%
+0.52	top10_5d	z=-0.07	J15=+0.46	-> spread_fwd_1d	avg=+0.6%
+0.51	book_1d	z=-0.09	J15=+1.26	-> soxs_fwd_5d	avg=-12.6%
+0.51	sc.bottom10_5d	z=-0.26	J15=+1.28	-> bottom10_fwd_5d	avg=+2.6%
+0.50	spy_vol_20d	z=+0.18	J15=+0.51	-> smh_fwd_1d	avg=+3.6%
+0.50	qqq_5d	z=-0.25	J15=+0.78	-> spread_fwd_1d	avg=+0.6%
-0.50	psq_5d	z=+0.27	J15=-0.82	-> spread_fwd_1d	avg=+0.6%

DOWN < -2.0% (n=183)

(none |r|>=0.5)

DOWN < -3.5% (n=56)

MODERATE (0.50-0.70):

-0.50	universe_20d	z=-0.37	J15=+0.28	-> spread_fwd_1d	avg=-0.3%
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Forward 5-Day portfolio return (filtered by book_fwd_5d)

UP > +4.5% (n=269)

(none |r|>=0.5)

UP > +6.5% (n=104)

MODERATE (0.50-0.70):

+0.51	sc.top10_5d	z=+0.10	J15=-1.03	-> spread_fwd_20d	avg=+1.8%
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DOWN < -4.5% (n=162)

(none |r|>=0.5)

DOWN < -6.5% (n=72)

MODERATE (0.50-0.70):

+0.58	sc.held_5d	z=+0.13	J15=-1.09	-> spread_fwd_20d	avg=-2.9%
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Forward 20-Day portfolio return (filtered by book_fwd_20d)

UP > +7.5% (n=422)

(none |r|>=0.5)

UP > +12.5% (n=209)

(none |r|>=0.5)

DOWN < -7.5% (n=161)

MODERATE (0.50-0.70):

+0.59	spy_20d	z=-0.04	J15=-0.18	-> spread_fwd_20d	avg=-2.5%
+0.54	universe_20d	z=-0.13	J15=+0.28	-> spread_fwd_20d	avg=-2.5%

DOWN < -12.5% (n=42)

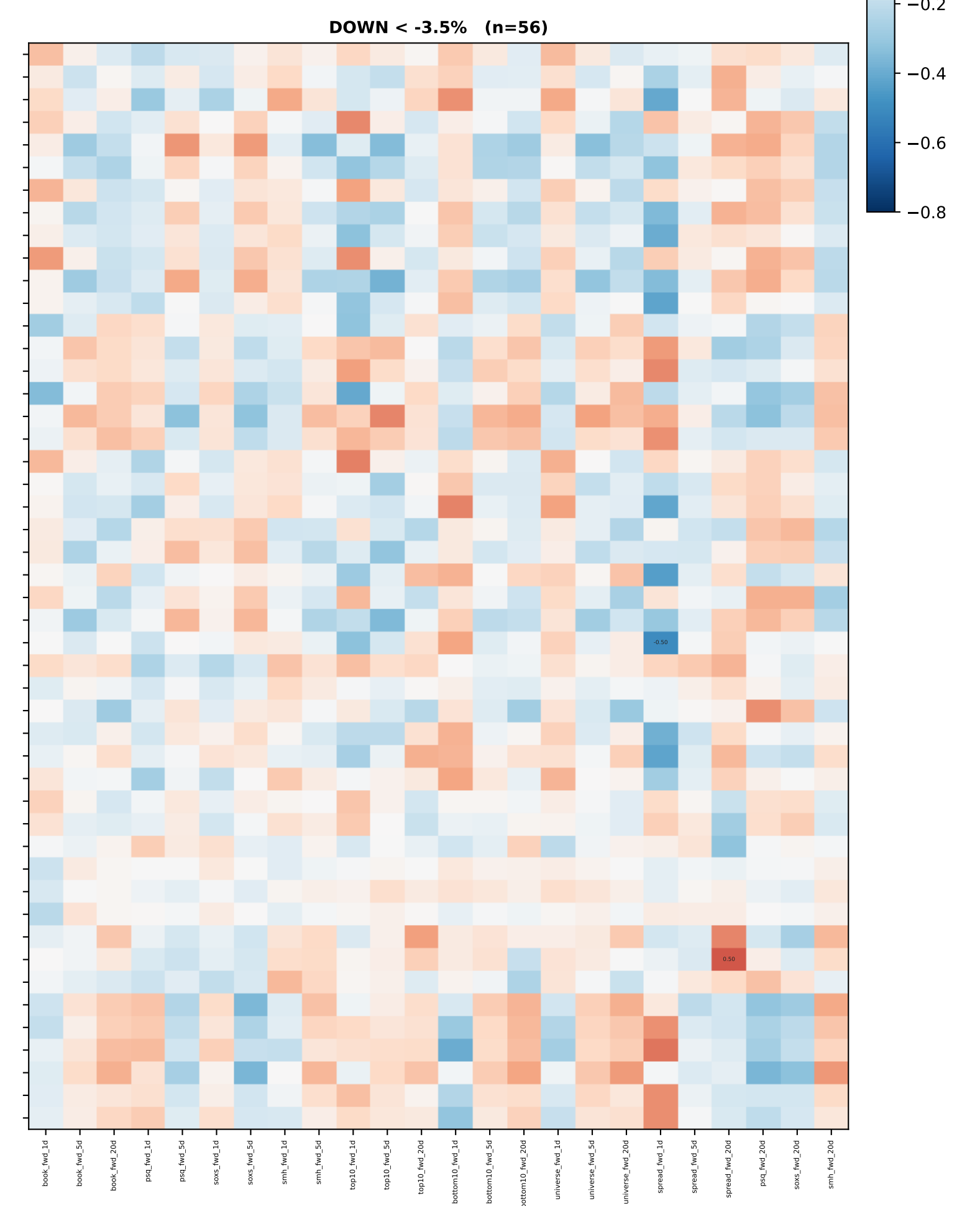
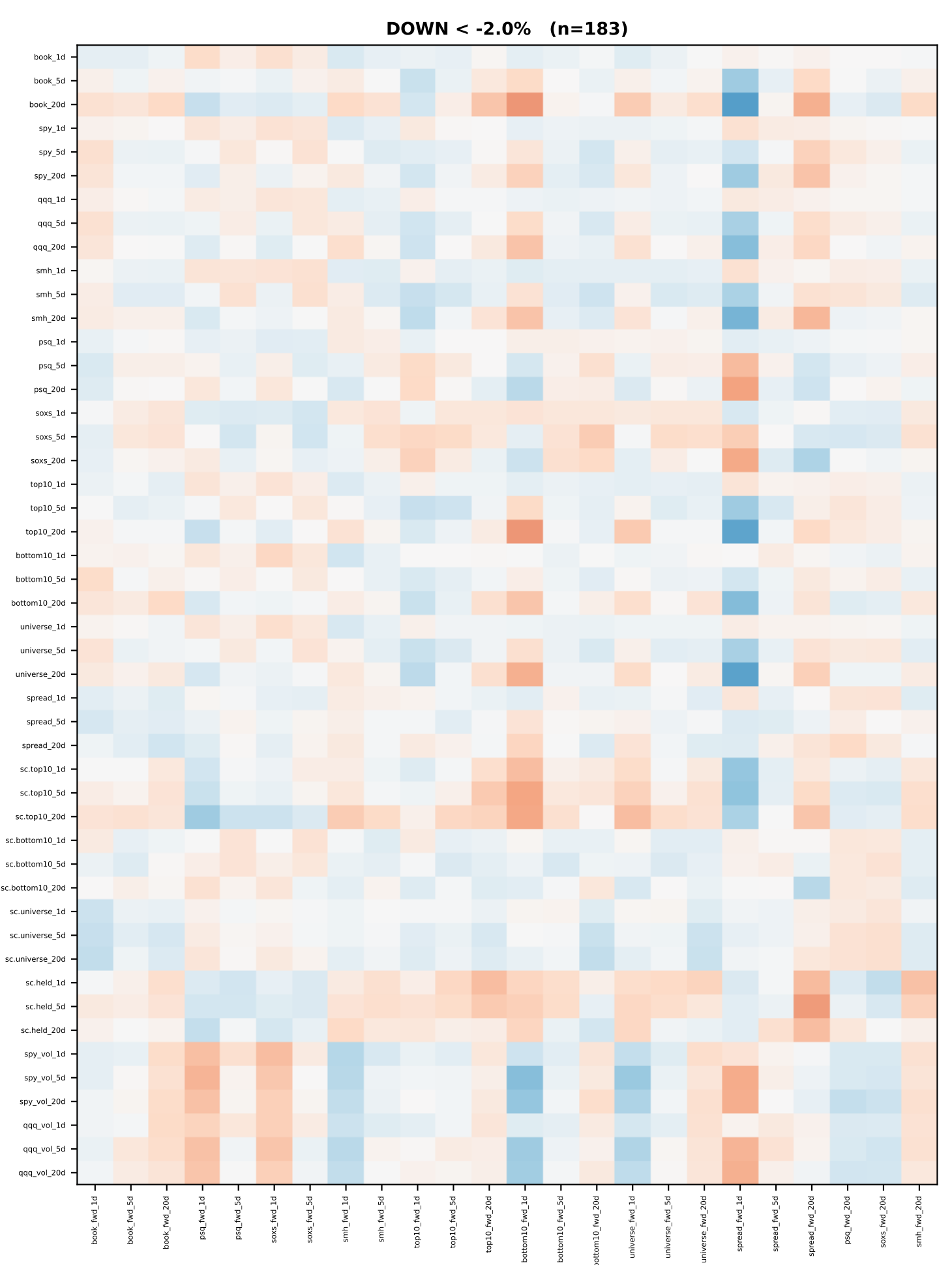
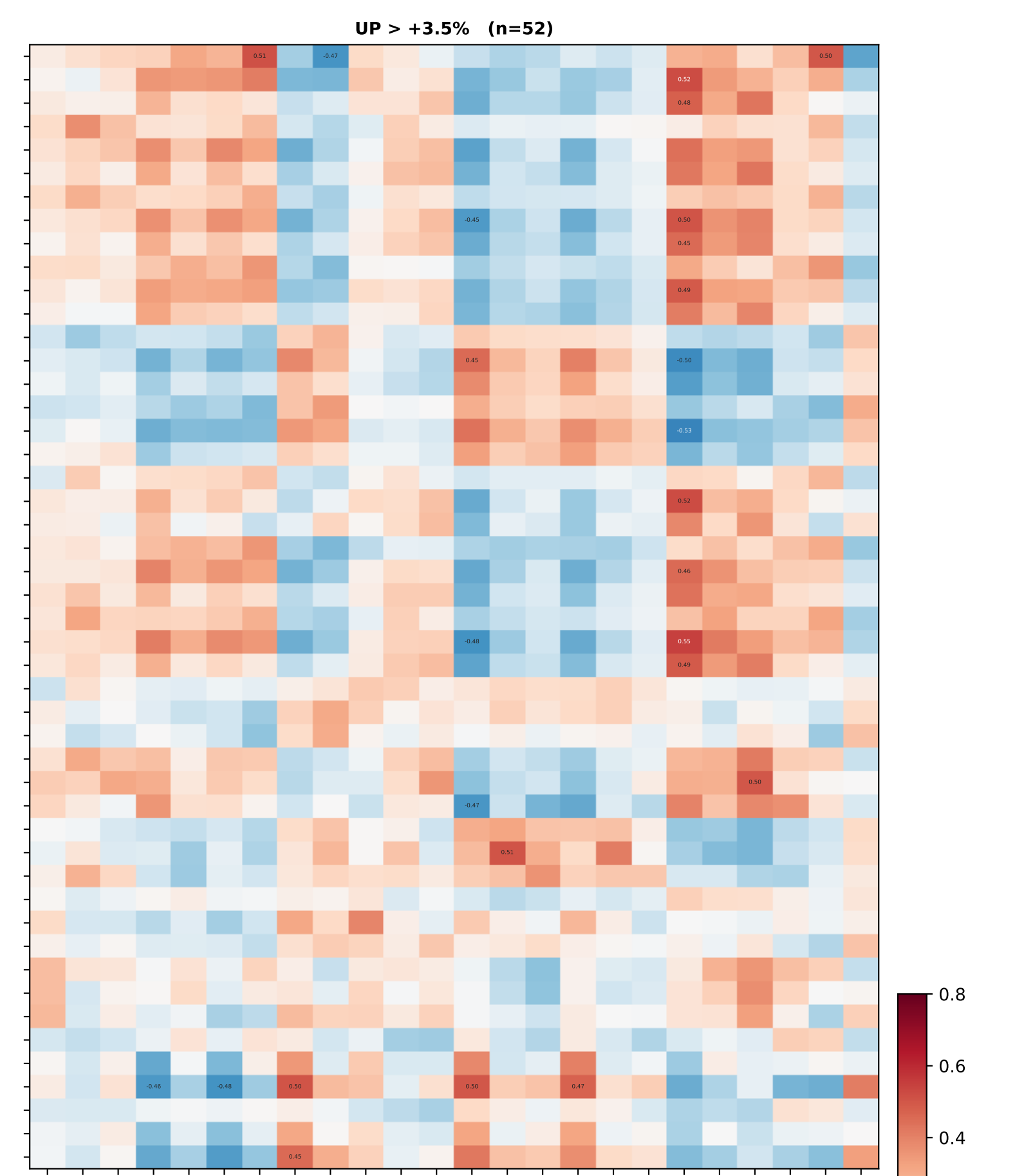
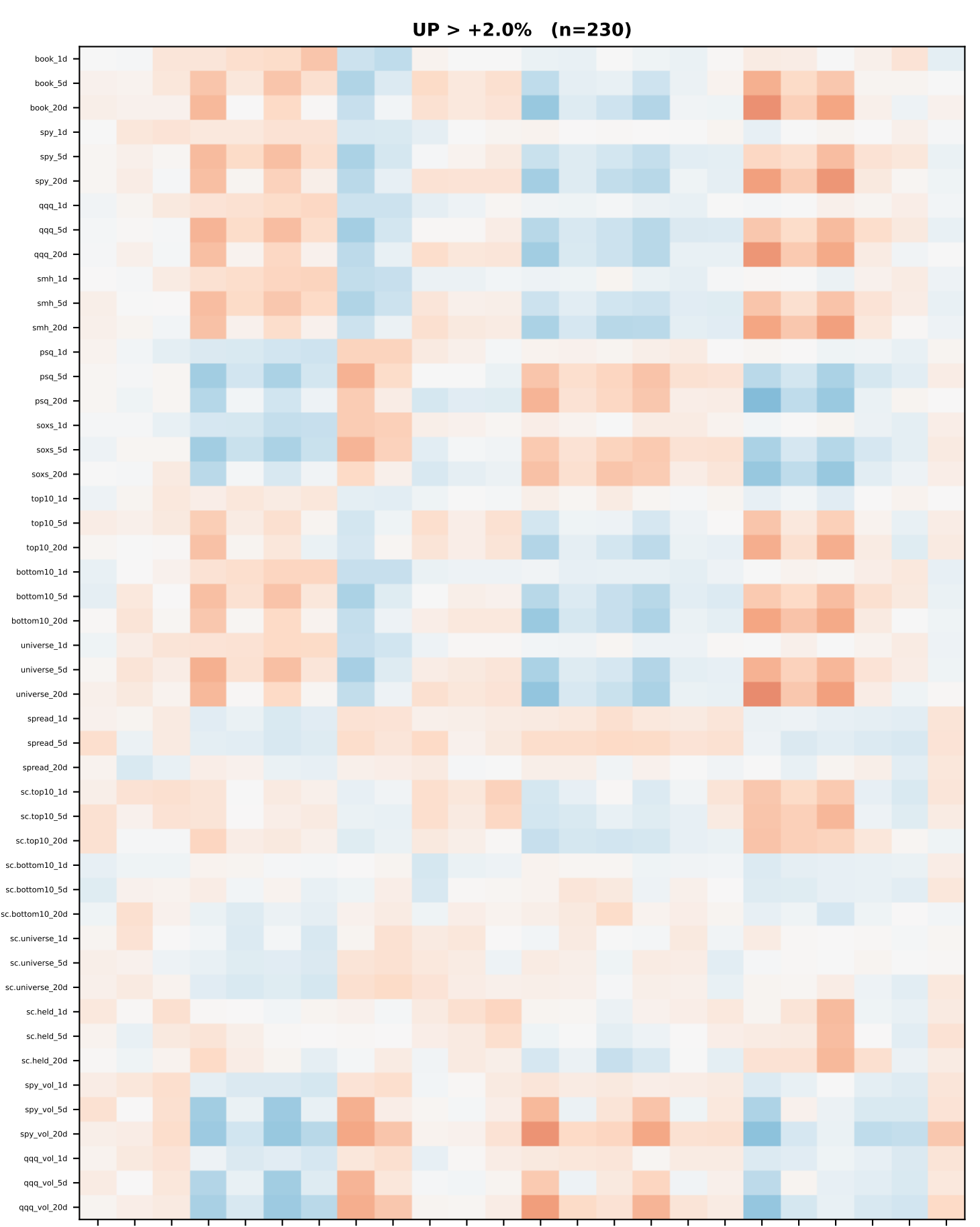
STRONG (|r|>=0.70):

+0.79	spy_20d	z=-0.33	J15=-0.18	-> spread_fwd_20d	avg=-2.4%
-0.71	psq_20d	z=+0.45	J15=+0.04	-> spread_fwd_20d	avg=-2.4%

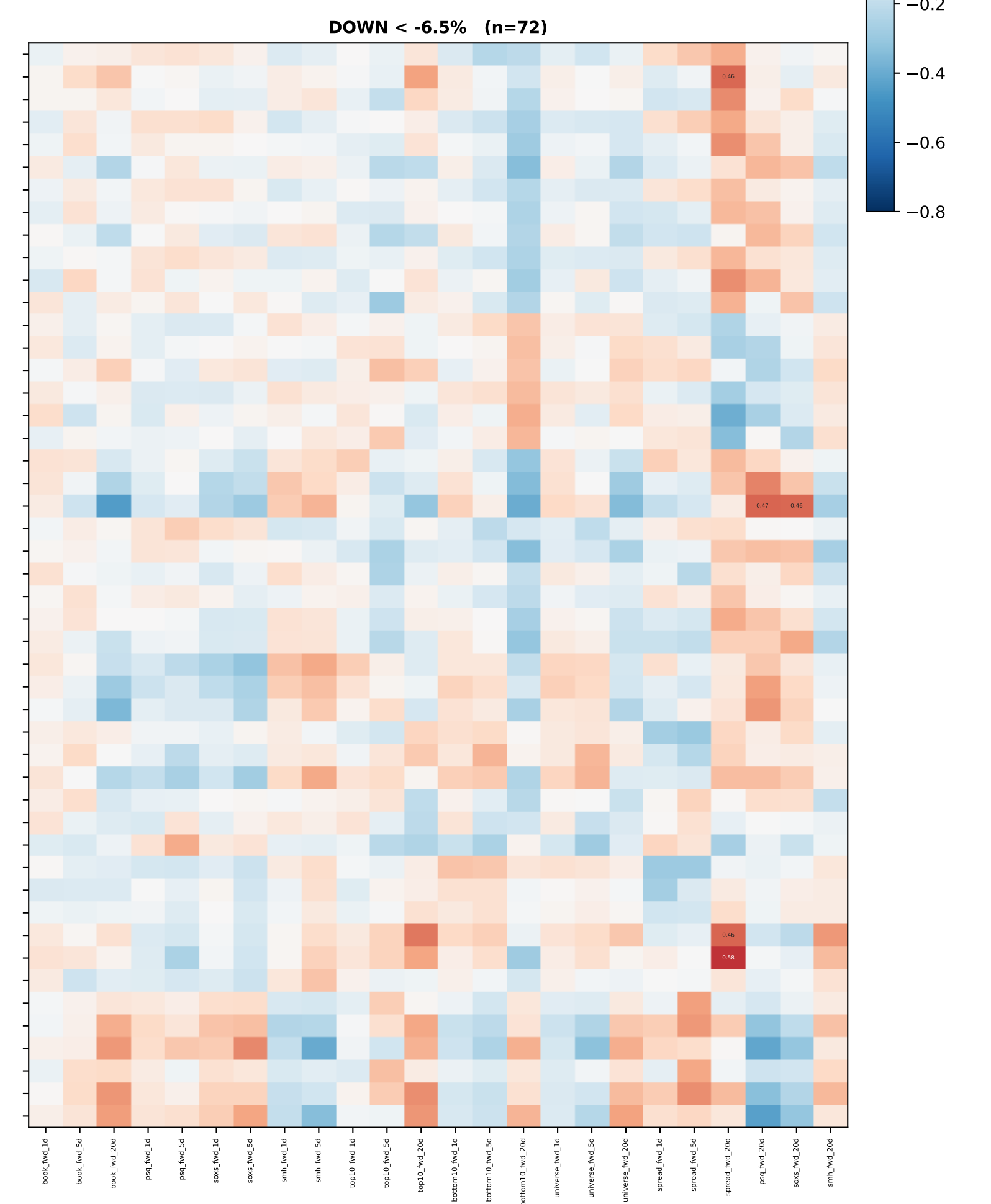
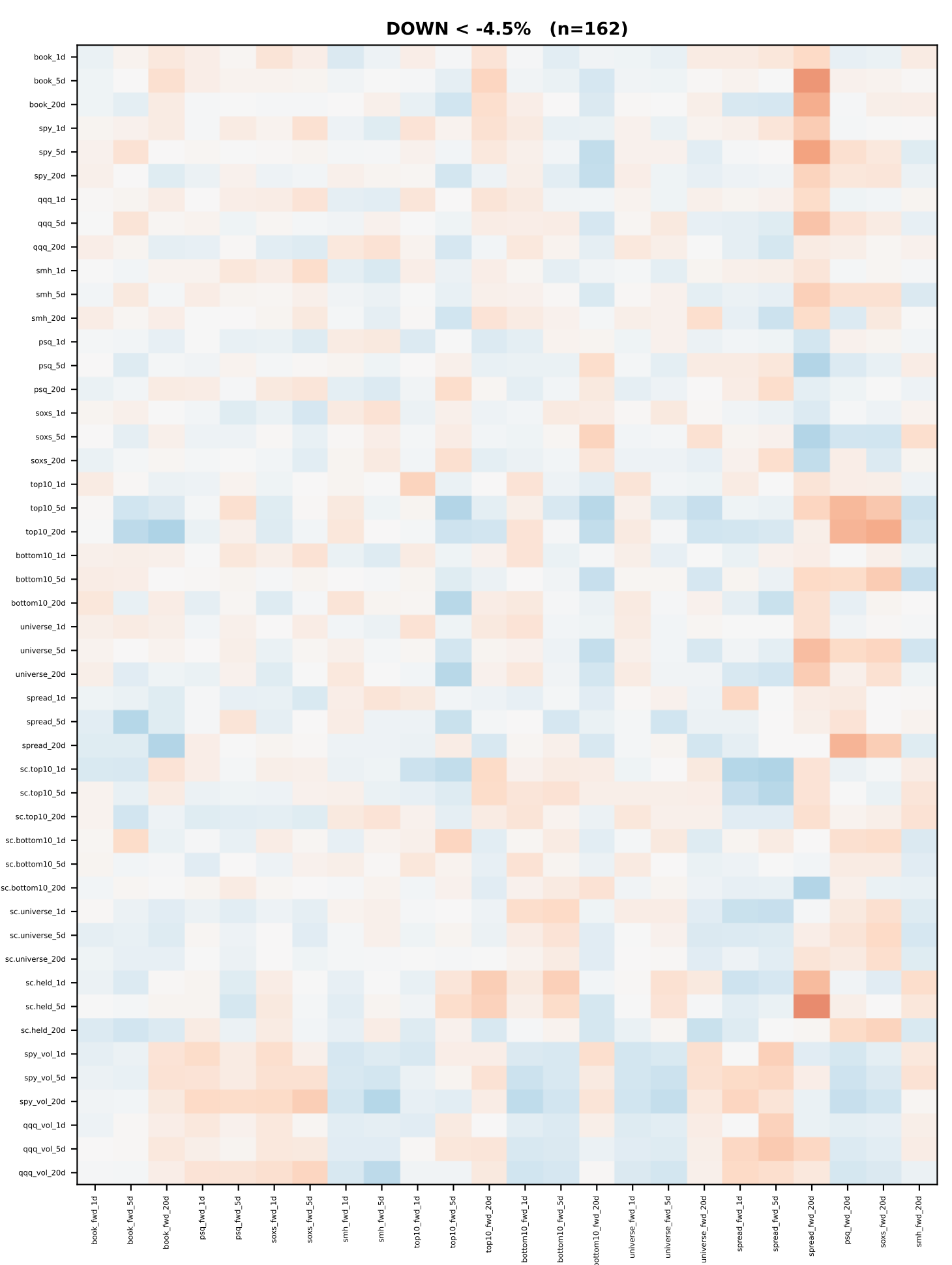
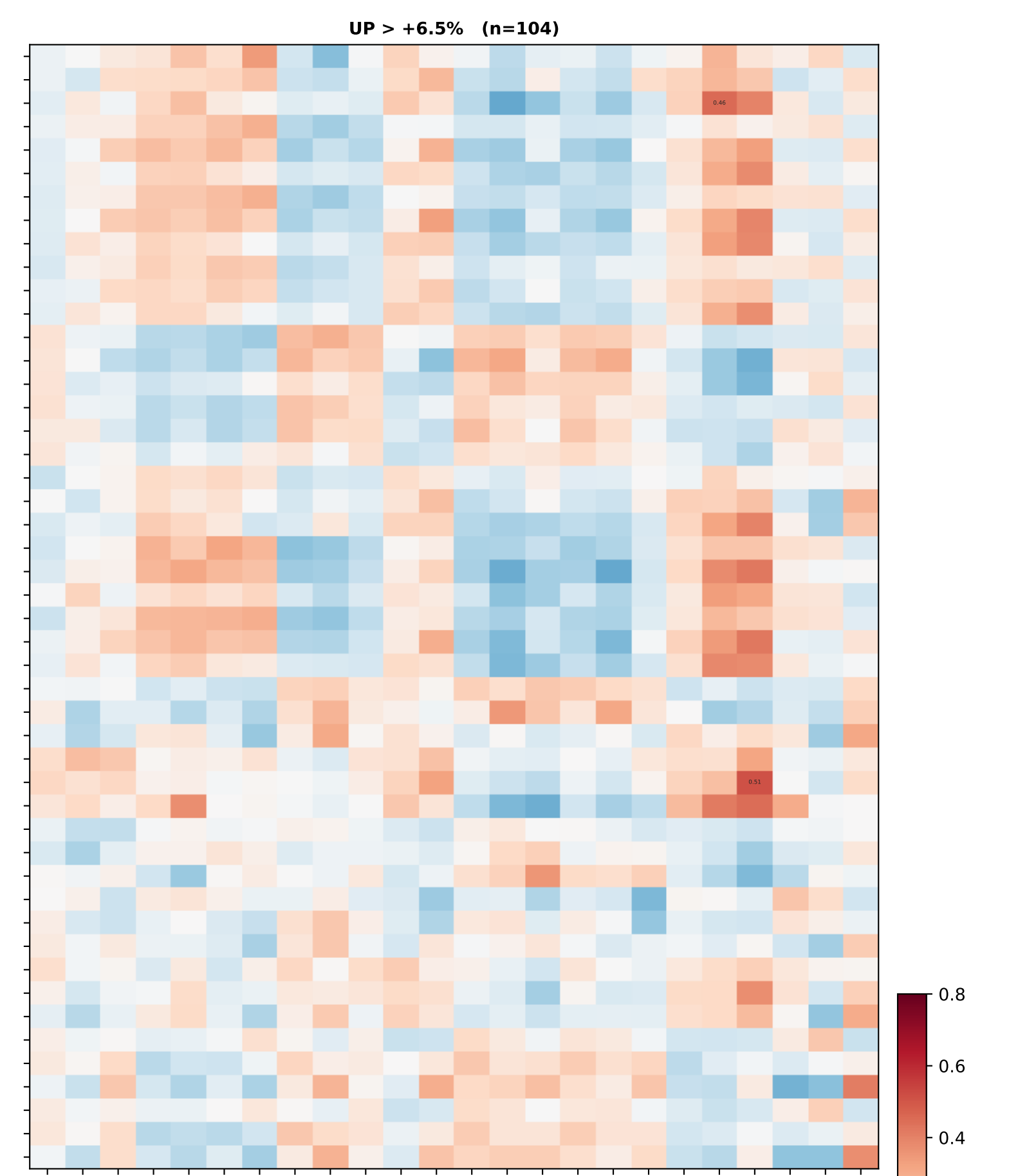
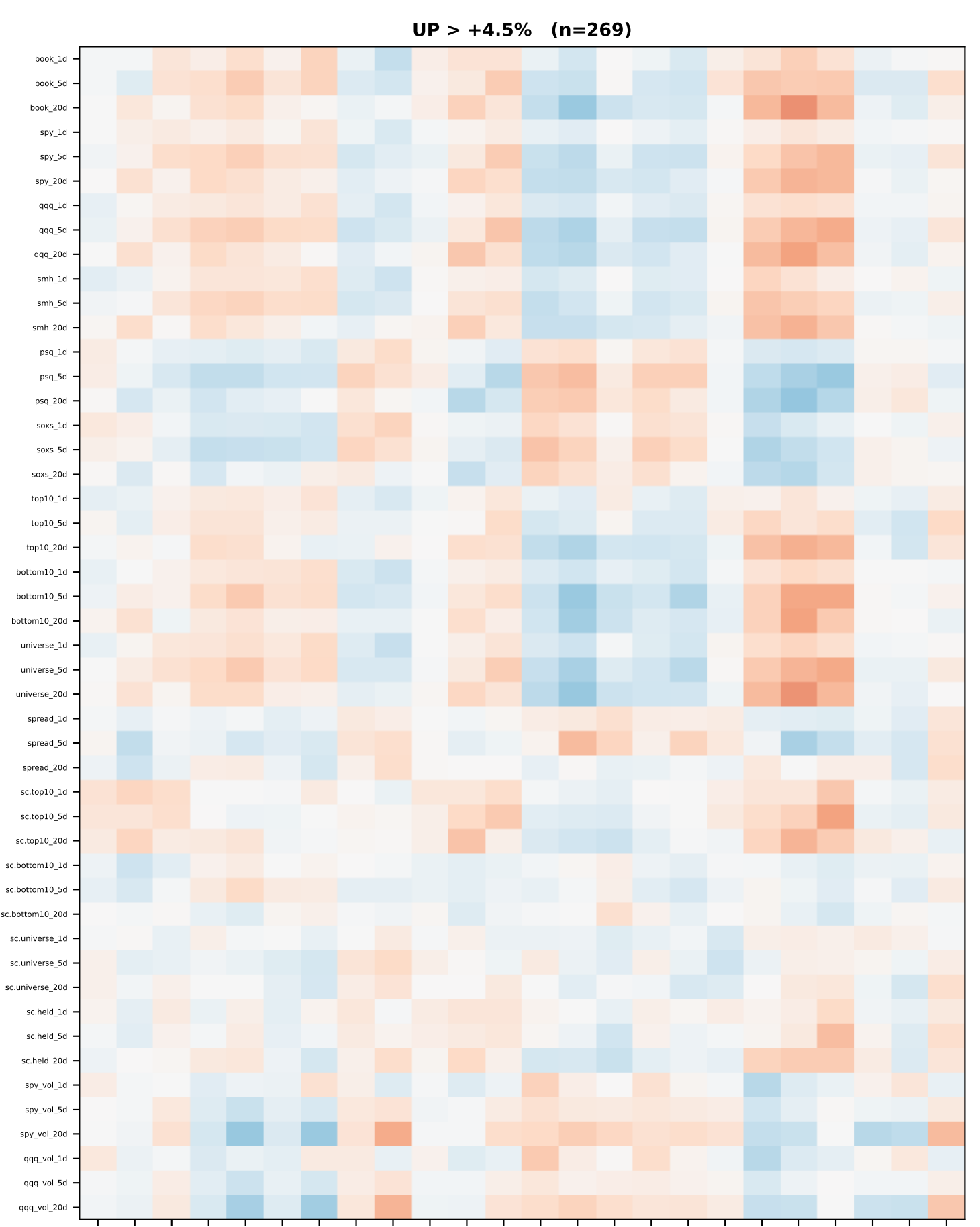
MODERATE (0.50-0.70):

+0.69	qqq_20d	z=-0.43	J15=-0.15	-> spread_fwd_20d	avg=-2.4%
-0.67	spy_20d	z=-0.33	J15=-0.18	-> bottom10_fwd_20d	avg=-12.2%
+0.66	universe_20d	z=-0.42	J15=+0.28	-> spread_fwd_20d	avg=-2.4%
+0.66	psq_5d	z=-0.05	J15=-0.82	-> bottom10_fwd_20d	avg=-12.2%
-0.66	qqq_5d	z=+0.05	J15=+0.78	-> bottom10_fwd_20d	avg=-12.2%
+0.66	book_5d	z=-0.12	J15=+0.49	-> spread_fwd_20d	avg=-2.4%
-0.66	sc.bottom10_20d	z=-0.07	J15=-0.38	-> spread_fwd_20d	avg=-2.4%
-0.64	sc.top10_20d	z=-0.25	J15=+0.23	-> book_fwd_20d	avg=-16.4%

+0.64	bottom10_20d	z=-0.47	J15=-0.22	-> spread_fwd_20d	avg=-2.4%
-0.64	soxs_20d	z=+0.53	J15=-0.71	-> spread_fwd_20d	avg=-2.4%
-0.63	sc.bottom10_20d	z=-0.07	J15=-0.38	-> psq_fwd_20d	avg=+9.9%
+0.62	spy_20d	z=-0.33	J15=-0.18	-> spread_fwd_5d	avg=+0.5%
+0.62	sc.bottom10_20d	z=-0.07	J15=-0.38	-> bottom10_fwd_20d	avg=-12.2%
+0.61	book_20d	z=-0.68	J15=-0.07	-> spread_fwd_20d	avg=-2.4%
-0.61	spy_5d	z=+0.05	J15=+0.61	-> bottom10_fwd_20d	avg=-12.2%
+0.61	qqq_5d	z=+0.05	J15=+0.78	-> spread_fwd_20d	avg=-2.4%
+0.61	smh_20d	z=-0.67	J15=+0.21	-> spread_fwd_20d	avg=-2.4%
-0.61	psq_5d	z=-0.05	J15=-0.82	-> spread_fwd_20d	avg=-2.4%
-0.61	book_5d	z=-0.12	J15=+0.49	-> bottom10_fwd_20d	avg=-12.2%
-0.60	smh_5d	z=+0.13	J15=+0.84	-> bottom10_fwd_20d	avg=-12.2%
-0.60	spread_5d	z=+0.21	J15=+0.70	-> book_fwd_20d	avg=-16.4%
-0.59	spy_vol_5d	z=+0.12	J15=+1.46	-> spread_fwd_20d	avg=-2.4%
-0.59	sc.top10_5d	z=-0.50	J15=-1.03	-> book_fwd_20d	avg=-16.4%
+0.58	soxs_5d	z=-0.11	J15=-1.07	-> universe_fwd_20d	avg=-12.2%
+0.57	spy_5d	z=+0.05	J15=+0.61	-> spread_fwd_20d	avg=-2.4%
-0.57	smh_5d	z=+0.13	J15=+0.84	-> universe_fwd_20d	avg=-12.2%
+0.57	psq_20d	z=+0.45	J15=+0.04	-> bottom10_fwd_20d	avg=-12.2%
+0.57	soxs_5d	z=-0.11	J15=-1.07	-> bottom10_fwd_20d	avg=-12.2%
-0.56	qqq_20d	z=-0.43	J15=-0.15	-> bottom10_fwd_20d	avg=-12.2%
-0.56	universe_5d	z=-0.03	J15=-0.10	-> bottom10_fwd_20d	avg=-12.2%
-0.56	qqq_vol_5d	z=+0.02	J15=+2.20	-> spread_fwd_20d	avg=-2.4%
+0.56	psq_5d	z=-0.05	J15=-0.82	-> universe_fwd_20d	avg=-12.2%
+0.55	bottom10_20d	z=-0.47	J15=-0.22	-> spread_fwd_5d	avg=+0.5%
-0.55	qqq_5d	z=+0.05	J15=+0.78	-> universe_fwd_20d	avg=-12.2%
+0.55	sc.bottom10_5d	z=+0.29	J15=+1.28	-> bottom10_fwd_20d	avg=-12.2%
-0.55	top10_5d	z=+0.27	J15=+0.46	-> book_fwd_20d	avg=-16.4%
+0.54	universe_5d	z=-0.03	J15=-0.10	-> spread_fwd_20d	avg=-2.4%
-0.54	sc.bottom10_20d	z=-0.07	J15=-0.38	-> soxs_fwd_20d	avg=+35.2%
+0.54	sc.bottom10_20d	z=-0.07	J15=-0.38	-> universe_fwd_20d	avg=-12.2%
+0.54	soxs_20d	z=+0.53	J15=-0.71	-> bottom10_fwd_20d	avg=-12.2%
-0.54	smh_20d	z=-0.67	J15=+0.21	-> bottom10_fwd_20d	avg=-12.2%
+0.53	smh_5d	z=+0.13	J15=+0.84	-> spread_fwd_20d	avg=-2.4%
+0.53	sc.held_20d	z=+0.02	J15=-0.00	-> smh_fwd_20d	avg=-12.6%
+0.53	spy_20d	z=-0.33	J15=-0.18	-> psq_fwd_20d	avg=+9.9%
+0.53	book_20d	z=-0.68	J15=-0.07	-> spread_fwd_5d	avg=+0.5%
-0.53	bottom10_20d	z=-0.47	J15=-0.22	-> book_fwd_20d	avg=-16.4%
-0.53	sc.bottom10_5d	z=+0.29	J15=+1.28	-> spread_fwd_20d	avg=-2.4%
+0.52	bottom10_20d	z=-0.47	J15=-0.22	-> top10_fwd_20d	avg=-14.6%
-0.52	psq_20d	z=+0.45	J15=+0.04	-> spread_fwd_5d	avg=+0.5%
+0.52	soxs_5d	z=-0.11	J15=-1.07	-> smh_fwd_20d	avg=-12.6%
-0.52	sc.bottom10_5d	z=+0.29	J15=+1.28	-> psq_fwd_20d	avg=+9.9%
-0.52	universe_20d	z=-0.42	J15=+0.28	-> bottom10_fwd_20d	avg=-12.2%
-0.51	spy_20d	z=-0.33	J15=-0.18	-> universe_fwd_20d	avg=-12.2%
+0.51	universe_20d	z=-0.42	J15=+0.28	-> spread_fwd_5d	avg=+0.5%
-0.51	spy_5d	z=+0.05	J15=+0.61	-> universe_fwd_20d	avg=-12.2%
+0.51	sc.bottom10_5d	z=+0.29	J15=+1.28	-> universe_fwd_20d	avg=-12.2%
+0.50	qqq_20d	z=-0.43	J15=-0.15	-> spread_fwd_5d	avg=+0.5%
-0.50	top10_5d	z=+0.27	J15=+0.46	-> bottom10_fwd_20d	avg=-12.2%
-0.50	top10_5d	z=+0.27	J15=+0.46	-> universe_fwd_20d	avg=-12.2%
+0.50	book_20d	z=-0.68	J15=-0.07	-> top10_fwd_20d	avg=-14.6%



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